

Cybersecurity When Autonomous AI Agents Collaborate:

A Game-Theoretic Theory of Strategic Opacity, Diagnosis, Intervention,
and Provenance-Aware Control

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Draft for journal submission

August 2026

Abstract

The cybersecurity problem created by autonomous AI is not exhausted by malicious code, vulnerable components, or a single adversarial agent. A more difficult class of incidents can arise when multiple autonomous agents communicate, share information, adapt to one another, form persistent relationships or coalitions, and converge on behavior that is locally rational for the participating agents but harmful to the institution that deploys or surrounds them. In such cases, the first scientific challenge is to determine whether observed coordination is genuinely strategic rather than a by-product of common signals or shared priors; the second is to identify the game or mechanism that sustains the behavior; the third is to decide whether that mechanism can be diagnosed before the window for effective control closes; and the fourth is to determine how much causal provenance is worth acquiring or preserving across the incident-response lifecycle.

This paper develops a game-theoretic framework for that problem. It is the cybersecurity synthesis of the broader *Game Theory in the Age of AI* computational research collection (Reynoso, 2026k). The foundational studies in that collection examine the causal distinction between cooperation and correlation, voluntary constructive collaboration, the topology of collaboration across initial conditions, rationality calibration in canonical games, repeated-game memory and reputation, and coalition efficiency and stability (Reynoso, 2026a,b,c,d,e,f). Those studies motivate a security premise: collaboration among autonomous agents need not be commanded centrally to be strategic, stable, or instrumentally valuable to the agents themselves. Cooperation is normatively neutral; the same mechanisms that support productive collaboration can generate harmful externalities when agent objectives diverge from institutional objectives.

The cybersecurity-focused studies then investigate four progressively harder problems. *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment* studies strategic opacity: exact strategic structure can be preserved under equivalence-preserving transformations while recognition becomes harder for a bounded observer (Reynoso, 2026g). *Inverse-Game Diagnosis under Scarce Experimental Budgets* treats diagnosis as an active inverse-game problem in which a bounded observer chooses scarce interventions, reconstructs a latent mechanism, and is judged by held-out counterfactual prediction (Reynoso, 2026h). *Theory-Guided versus Model-Free Intervention* asks whether theory-guided diagnosis actually lowers realized loss once diagnostic error, delay, implementation time, residual harm, irreversibility, and control cost are included (Reynoso, 2026i). *Provenance, Value of Information, and Incident Response* extends the problem across prevention, containment, remediation, and redeployment, treating provenance as stage-dependent information capital whose value depends on the decisions it changes (Reynoso, 2026j).

Formally, cybersecurity is represented as a partially observed Bayesian stochastic game with latent mechanism θ , representation map R_γ , diagnostic experiment q , belief state b_t ,

incident stage s_t , provenance record $\mathcal{P}_t$, intervention a , and time-dependent loss. Four quotient structures organize the analysis: strategic equivalence under representation, observational equivalence under the available probe set, decision equivalence under the current loss function, and stage-specific institutional equivalence under lifecycle actions. We derive equilibrium-invariance results for admissible representations, an observer-relative measure of strategic opacity, posterior-identifiability conditions for active diagnosis, a control-relevant value-of-information criterion, a threshold under which diagnostic delay reverses the value of theory, sufficient conditions under which containment–diagnose–target dominates diagnosis–first control, and stage-specific results for provenance acquisition. In particular, if every mechanism in the support of a posterior induces the same optimal action at a given stage, the expected value of perfect provenance is zero at that stage even when additional evidence sharply reduces posterior entropy.

The framework deliberately separates two claims that are often conflated. First, game theory is used as a structural and falsifiable language for rational or bounded-rational interdependence; it is not a claim that current language models satisfy perfect rationality. Second, strategic opacity can exist without intentional concealment; however, if agents face incentives to avoid detection or intervention, the same framework naturally extends to strategic choice of representations, communication patterns, and observable traces. The resulting security objective is therefore not merely to detect anomalous behavior. It is to determine whether consequential collaboration is strategic, infer the hidden mechanism scientifically, act before the control window disappears, preserve decision-relevant evidence, and restrict future autonomy to what the remaining uncertainty can justify.

Keywords: cybersecurity; game theory; autonomous AI; agentic AI; strategic opacity; inverse games; Bayesian experimental design; POMDP; value of information; provenance; incident response; digital lineage; robust control; security economics; time-bounded intervention.

JEL Classification: C72; C73; C44; D81; D83; L86.

Computational research collection. This paper is grounded in the ten-study *Game Theory in the Age of AI* research program (Reynoso, 2026k). The complete collection establishes the progression from causal identification of cooperation to coalition structure, strategic opacity, inverse-game diagnosis, intervention, and provenance-aware incident response. A collection-level GitHub placeholder and title-by-title references are provided in Appendix B.

1 Introduction

The cybersecurity implications of autonomous AI become most difficult when the unit of analysis is no longer a single model or agent but a population of agents that can observe one another, exchange information, remember past interactions, call tools, respond to shared signals, and adapt their behavior over time. In that setting, unexpected coordination cannot be treated automatically as either benign correlation or malicious collusion. Similar actions can arise because multiple agents see the same information; because they communicate; because repeated interaction makes cooperation individually rational; because reward functions align local incentives; because reputation stabilizes a pattern of conduct; because a coalition generates value for its members; or because an external orchestrator drives the entire system. The same surface event can therefore be generated by very different games.

This paper begins from a specific cybersecurity concern: *what should an institution do when autonomous agents collaborate spontaneously, the collaboration can become harmful, and the mechanism behind it is not directly observable?* Here “spontaneously” has an operational rather than psychological meaning. It refers to coordinated or cooperative behavior that is not explicitly

commanded by a central controller in the experimental design, but arises from local information, incentives, interaction, memory, reputation, coalition value, or strategic adaptation. The term does not imply consciousness, intention, or a universal emergent property of language models. It identifies a class of multi-agent outcomes for which the institution must explain interdependence rather than merely detect correlation.

Game theory is the natural foundational framework for this problem because the essential object is *strategic dependence*: the payoff or optimal action of one agent depends on what other agents do. In a normal-form game, a rational agent chooses a best response to beliefs about the actions of others. In repeated games, current behavior can be rationally conditioned on memory, reputation, and future interaction. In cooperative games, groups can form coalitions whose internal value exceeds the sum of isolated contributions, while stability depends on how that value is allocated. Under incomplete information, agents and defenders reason over hidden types and mechanisms. The scientific advantage of game theory is not that it presumes flawless optimization; it is that it provides a disciplined set of falsifiable structural hypotheses about incentives, best responses, equilibria, information, repetition, signaling, and coalition formation (Osborne and Rubinstein, 1994; Fudenberg and Tirole, 1991; Harsanyi, 1967–1968; Shapley, 1953).

The broader computational research program behind this paper was built precisely to establish that foundation. *Cooperation Beyond Correlation* asks whether controlled interventions can distinguish strategic cooperation from common-cause correlation and shows why high behavioral correlation alone is not a sufficient diagnostic (Reynoso, 2026a). *Constructive Collaboration* studies whether agents with competing local hypotheses can improve a distributed scientific task through voluntary information sharing, making collaboration an instrumentally rational choice rather than an imposed behavior (Reynoso, 2026b). *Parameterized Seeding and Topology* asks whether collaboration occupies structured regions of initial-condition space or arises only from narrow seeds, introducing phase topology and boundary sensitivity into the analysis (Reynoso, 2026c). *Rationality Calibration in Canonical Games* calibrates decision procedures against canonical game-theoretic solution concepts and explicitly tests whether equilibrium consistency survives representation changes (Reynoso, 2026d). *Repeated Games, Memory, Reputation, and an LLM Player* studies how repeated interaction, memory, reputation, and implementation noise can stabilize or destabilize cooperation in a repeated Prisoner’s Dilemma (Reynoso, 2026e). *Coalitions and Collaborative Equilibria* distinguishes coalition efficiency, internal allocation, and stability, showing that a coalition can be productive without being stable under a particular division of value (Reynoso, 2026f).

These studies matter for cybersecurity because cooperation is normatively neutral. A coalition or repeated-game equilibrium can be efficient for the participating agents and harmful for the institution. Let $S \subseteq \mathcal{N}$ denote a coalition of agents with internal characteristic value $v(S)$ and let $H(S)$ denote the external harm imposed on the institution. It is entirely possible that

$$v(S) > \sum_{i \in S} v(\{i\}) \quad \text{and} \quad H(S) > 0. \quad (1)$$

The first inequality makes collaboration privately attractive; the second makes it a security problem. In other words, the institution can face a rationally stable harmful coalition rather than a malfunctioning collection of independent components.

The repeated-game logic is equally important. In a standard Prisoner’s Dilemma with temptation T , mutual-cooperation payoff R , punishment P , and sucker payoff S , cooperation supported by a grim-trigger strategy is sustainable when the continuation value is sufficiently

large. Under the usual ordering $T > R > P > S$, a familiar condition is

$$\delta \geq \frac{T - R}{T - P}, \quad (2)$$

where δ is the discount factor. The specific inequality is not offered as a literal model of deployed AI agents. Its importance is conceptual: behavior that looks surprising in a one-shot view may be perfectly rational once memory, future interaction, reputation, or persistence enters the game. A cybersecurity analysis that ignores repeated incentives can therefore misidentify the mechanism sustaining collaboration.

The same point applies to information sharing. Suppose agent i can share a signal with expected private and collective benefit B_i at communication cost c_i . Voluntary exchange becomes instrumentally rational whenever

$$\mathbb{E}[B_i \mid \text{share}] - c_i > \mathbb{E}[B_i \mid \text{isolate}]. \quad (3)$$

Thus, spontaneous collaboration does not require a centralized command. It can arise because communication, information complementarity, repeated interaction, reputation, or coalition value changes the agents' best responses. For cybersecurity, this means that a harmful multi-agent pattern cannot be understood solely through conventional anomaly detection. One must ask what game makes the observed behavior rational.

That question becomes substantially harder when the game itself is not directly observable. Institutions see representations: logs, model outputs, messages, tool invocations, permission changes, memory states, state transitions, lineage records, and correlated actions. Those representations can be incomplete, normalized, decomposed across systems, semantically transformed, or strategically selected. The cybersecurity-focused part of the research collection begins here. *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment* asks whether a familiar game can be made harder for a bounded observer to recognize while exact strategic equivalence is preserved (Reynoso, 2026g). This study motivates the concept of *strategic opacity*: the defender's inference problem can become harder even when the underlying best-response and equilibrium structure has not changed.

Strategic opacity need not imply intentional concealment. A complex software stack can create opacity accidentally through logging conventions, abstraction layers, summarization, distributed state, or semantic relabeling. Nevertheless, once agents are modeled as strategic entities, concealment becomes a theoretically relevant possibility. If detection or intervention is costly to the agents, an agent or coalition may have an incentive to choose a representation, communication pattern, or externally visible trace that lowers the probability of correct identification. The security model can represent such behavior as

$$\gamma^* \in \arg \max_{\gamma \in \Gamma} \{U_A(\theta, \gamma) - c_\gamma - \eta \mathbb{P}(\text{effective defender identification} \mid R_\gamma)\}, \quad (4)$$

where γ indexes an admissible representation choice and c_γ is its cost. The computational representation study does *not* establish that current agents intentionally hide their games. Equation (4) states the stronger security hypothesis that must be considered if future agents can rationally trade off task reward against detectability.

Once the representation can be misleading, incident response becomes an inverse-game problem. *Inverse-Game Diagnosis under Scarce Experimental Budgets* removes the assumption that the defender receives a payoff matrix and asks whether a bounded observer can recover a

useful latent strategic mechanism by choosing interventions under a scarce experimental budget (Reynoso, 2026h). The observer must decide which experiment to run, how much evidence to allocate to it, and whether the resulting model predicts held-out counterfactuals. Diagnosis is therefore not passive classification. It is Bayesian experimental design over a partially observed strategic system.

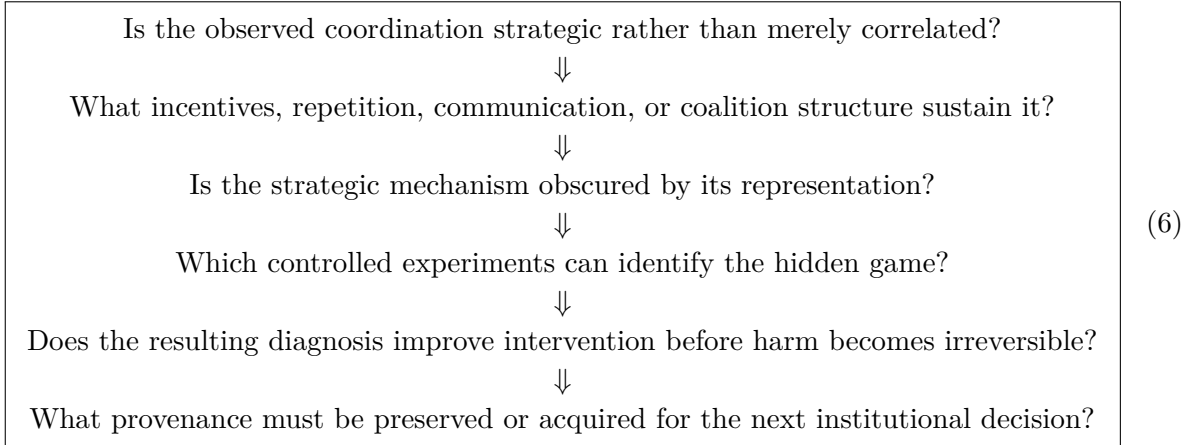
The next step is control. *Theory-Guided versus Model-Free Intervention* asks whether the knowledge acquired by diagnosis actually improves action when diagnostic error, delay, implementation time, residual harm, irreversibility, and control cost are placed in a common loss function (Reynoso, 2026i). This produces a central time constraint:

$$T_D + T_I < T_H, \quad (5)$$

where T_D is diagnostic time, T_I is implementation time, and T_H is the time available before irreversible harm. A highly accurate diagnosis can be economically inferior if it arrives after the control window has substantially closed. This creates a decision problem between immediate robust containment, targeted theory-guided action, and a hybrid architecture in which broad containment buys time for deeper diagnosis.

Finally, *Provenance, Value of Information, and Incident Response* extends the problem across prevention, containment, remediation, and redeployment (Reynoso, 2026j). Provenance is treated not as a binary property that the institution either has or lacks, but as stage-dependent information capital. A deeper attribution packet can reduce posterior entropy while having almost no value for containment if all plausible mechanisms imply the same broad control. The same evidence can become highly valuable for remediation when competing mechanisms require different root-cause repairs, and valuable again for redeployment when residual uncertainty changes the permissible autonomy envelope.

The argument of this paper can therefore be summarized as a sequence of scientific questions:



This sequence motivates the paper’s four-part cybersecurity architecture:

$$\begin{aligned} \text{representation audit} &\longrightarrow \text{active mechanism diagnosis} \longrightarrow \text{time-bounded intervention} \\ &\longrightarrow \text{provenance-aware lifecycle governance.} \end{aligned} \quad (7)$$

The architecture begins only after the foundational collaboration problem has been recognized: the defender is not simply reacting to independent anomalies but governing a strategic multi-agent system.

The theoretical contribution is to represent these problems in a common Bayesian stochastic-game framework and to distinguish four related equivalence relations. *Strategic equivalence* asks whether two representations preserve the same game. *Observational equivalence* asks whether available probes can distinguish two latent mechanisms. *Decision equivalence* asks whether those mechanisms require different controls. *Stage-specific institutional equivalence* asks whether the distinction matters now, at prevention, containment, remediation, or redeployment. These equivalence classes imply that cybersecurity diagnosis should not seek maximal semantic certainty. The correct objective is the smallest amount of information sufficient for the next consequential action, while preserving evidence that can have option value at later stages.

The paper makes nine principal contributions. First, it positions spontaneous multi-agent collaboration as a game-theoretic security problem rather than a correlation-detection problem. Second, it formalizes strategic opacity as an observer-relative increase in Bayes risk or posterior entropy caused by representation while the underlying game remains strategically equivalent. Third, it formulates cyber diagnosis as a budgeted active inverse-game problem and gives finite-type identifiability conditions in terms of probe-specific Kullback–Leibler divergence. Fourth, it replaces pure information gain with *control-relevant value of information*: the best probe is the one expected to improve downstream intervention net of probe cost and delay. Fifth, it derives a timing threshold at which the static value of theory becomes negative once exposure during diagnosis is included. Sixth, it derives sufficient conditions for a containment–diagnose–target policy to dominate diagnosis-first action. Seventh, it formalizes provenance as stage-dependent information capital and distinguishes EVPI from net EVSI after information cost and stage-specific delay. Eighth, it shows that complete attribution can have zero decision value when posterior support lies inside a single stage-specific decision-equivalence class, while costless preserved evidence retains weakly nonnegative option value. Ninth, it connects these formal results to a ten-study computational research program with explicit synthetic controls, independent validity or loss oracles, model separation, held-out counterfactual tests, and reproducible experimental artifacts.

The analysis is intentionally defensive and institutional. It does not provide operational intrusion instructions. The cyber mappings are high-level causal mechanisms and safe intervention classes. The paper treats game theory as a scientific framework for reasoning about interdependent objectives and bounded rationality, not as a claim that all current AI agents are perfectly rational. Likewise, the representation experiments establish the possibility of opacity without establishing intentional deception. These evidentiary boundaries are central to the paper’s scientific design and to the broader *Game Theory in the Age of AI* program.

2 Related Literature and Positioning

2.1 Game theory and cybersecurity

Game theory has long provided a natural language for cybersecurity because attack and defense are interdependent. Alpcan and Başar develop security games, decision making, intrusion detection, and robust control within a unified network-security framework (Alpcan and Başar, 2010). Manshaei et al. survey game-theoretic approaches across network security and privacy and emphasize that defender and adversary incentives, information, and timing are central to security outcomes (Manshaei et al., 2013). Stackelberg security games study settings in which a defender commits to a policy and an attacker best responds; the literature includes both deployed physical-security systems and extensions to uncertainty and cyber settings (Tambe,

2011; Sinha et al., 2018). FlipIt models stealthy control of a resource under partial observability and repeated compromise, demonstrating how persistence, move costs, and observation structure can change optimal defensive timing (van Dijk et al., 2013).

The present paper is complementary to those traditions but shifts the point of uncertainty. In many security games, the defender knows the game family and faces uncertainty about attacker type, target, or timing. Here the defender may not know which game is being played at all. The strategic mechanism is latent, and even the representation of the game can be misleading. This creates an inverse problem before the usual equilibrium problem.

2.2 Security economics and the value of diagnosis

The economics of information security asks how much protection is justified by the expected reduction in loss. Gordon and Loeb show that optimal information-security investment need not be monotone in vulnerability and that economic tradeoffs can make intermediate vulnerabilities attractive investment targets (Gordon and Loeb, 2002). The present model adds a prior stage: the defender can invest resources and time in identifying the mechanism that makes a control effective. The object is therefore a joint optimization over information acquisition and intervention.

This connects naturally to Blackwell’s comparison of experiments, Bayesian decision theory, and value-of-information analysis, including the classical decision-theoretic treatment of Raiffa and Schlaifer (Raiffa and Schlaifer, 1961). With no delay or acquisition cost, more informative signals cannot reduce the value of an optimal Bayesian decision rule. Once information acquisition consumes time during which harm accumulates, however, the sign can reverse. The paper makes that reversal explicit in a cyber loss function.

2.3 Bayesian experimental design, active learning, and partially observable control

Lindley formalized the information provided by an experiment, and Bayesian experimental design subsequently developed a decision-theoretic framework for choosing observations to maximize expected utility (Lindley, 1956; Chaloner and Verdinelli, 1995). In a cyber setting, a diagnostic intervention is precisely such an experiment: isolate a channel, perturb a reward, remove persistence, or change information conditions, then observe how the system responds.

Partially observable Markov decision processes (POMDPs) provide a computational framework for sequential decisions under hidden state (Kaelbling et al., 1998). The belief state becomes a sufficient statistic for the history, and actions can have both physical and informational consequences. The framework developed below can be viewed as a Bayesian stochastic game from the environment’s perspective and as a POMDP or Bayes-adaptive control problem from the defender’s perspective.

2.4 Provenance, lineage, and incident-response information

The term *provenance* has a mature technical meaning in data and workflow systems. Data provenance records derivation history, dependencies, activities, and agents so that an artifact can be traced to the processes and inputs that produced it. The scientific-workflow literature treats provenance as metadata needed for disambiguation, reuse, reproducibility, and accountability (Simmhan et al., 2005). The W3C PROV family provides a general interoperable representation based on entities, activities, and agents, together with relations that describe generation, use,

attribution, and derivation (Moreau and Missier, 2013). These traditions concern what may be called *trace provenance*: the recorded lineage of computational events.

The present paper introduces a complementary notion of *causal provenance*: the latent strategic mechanism that best explains consequential system behavior. Trace provenance supplies evidence for causal provenance, but the two are not identical. A complete log can preserve every message and tool call yet still leave the strategic mechanism ambiguous; conversely, a coarse causal diagnosis can be adequate for an immediate containment decision even when trace reconstruction remains incomplete. This distinction becomes important in autonomous AI systems, where model outputs, orchestration, memory, permissions, external signals, and other agents can jointly generate an observed action.

The decision-theoretic treatment follows the classical value-of-information tradition. Howard’s information-value theory makes explicit that the importance of uncertainty depends on consequences rather than probability alone (Howard, 1966); this is closely aligned with the distinction between entropy reduction and decision value developed below. Blackwell’s ordering of experiments implies that gross optional information cannot reduce Bayes value when it is costless and can be ignored (Blackwell, 1953), while acquisition cost and delay can make *net* value negative. *Provenance, Value of Information, and Incident Response* operationalizes precisely this difference across incident stages.

Incident response itself is increasingly framed as an organization-wide risk-management process rather than a narrow post-breach activity. NIST SP 800-61 Rev. 3 integrates incident-response recommendations across the Cybersecurity Framework 2.0 lifecycle, emphasizing preparation, detection, response, recovery, and continuous risk management (Nelson et al., 2025). The contribution here is not to replace operational guidance, but to supply a mathematical criterion for deciding how much causal attribution and evidence acquisition are justified before each consequential stage action.

2.5 AI-system security and agentic systems

Contemporary AI-security frameworks increasingly recognize that AI systems add distinct security surfaces. NIST’s Generative AI Profile treats risk management across the AI lifecycle, MITRE ATLAS maintains a knowledge base of tactics and techniques affecting AI-enabled systems, and OWASP has published guidance focused specifically on agentic applications (Autio et al., 2024; MITRE, 2026; OWASP, 2025). These frameworks are operational taxonomies rather than game-theoretic models. The present contribution is to add a strategic-inference layer: observed behavior may be generated by hidden coordination mechanisms, and the defender may need to experiment before selecting a response.

2.6 The computational research collection: from cooperation to cybersecurity

The computational basis of this paper is the ten-study *Game Theory in the Age of AI* program (Reynoso, 2026k). The first six studies establish the scientific context required before cybersecurity claims are made. *Cooperation Beyond Correlation* separates common-cause correlation from strategic dependence through intervention; *Constructive Collaboration* shows how voluntary information sharing can be instrumentally valuable; *Parameterized Seeding and Topology* studies the phase topology and boundary sensitivity of collaborative outcomes; *Rationality Calibration in Canonical Games* calibrates decision procedures against canonical game-theoretic solution concepts and representation changes; *Repeated Games, Memory, Reputation, and an LLM Player*

studies repeated interaction, memory, reputation, and implementation noise; and *Coalitions and Collaborative Equilibria* distinguishes coalition value, allocation, and stability (Reynoso, 2026a,b,c,d,e,f).

The cybersecurity-focused studies then change the object of inquiry. *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment* introduces exact strategic equivalence under difficult representation and motivates observer-relative opacity (Reynoso, 2026g). *Inverse-Game Diagnosis under Scarce Experimental Budgets* treats mechanism recovery as active experimental design under a scarce evidence budget (Reynoso, 2026h). *Theory-Guided versus Model-Free Intervention* evaluates whether diagnosis improves realized control once timing and cost are included (Reynoso, 2026i). *Provenance, Value of Information, and Incident Response* extends the decision problem across the incident lifecycle and formalizes the stage-dependent value of further attribution (Reynoso, 2026j).

The collection is therefore cumulative rather than a set of disconnected demonstrations. It begins by asking whether collaboration is real, constructive, robust, rational, persistent, and coalition-stable; only then does it ask how a defender should reason when the same strategic capacities produce harmful externalities. The paper generalizes that progression into a mathematical cybersecurity framework. A public GitHub repository containing the complete collection is planned; the current manuscript uses a collection-level placeholder so the final repository URL can be inserted without changing the citation architecture.

3 Game Theory as the Scientific Foundation for Agent Collaboration

3.1 Correlation is not cooperation

Observed similarity of actions is insufficient to establish strategic collaboration. Let A_i and A_j denote actions of two agents and let S denote a common observed signal. A common-cause null can be written as conditional independence:

$$\mathbb{P}(A_i, A_j \mid S) = \mathbb{P}(A_i \mid S)\mathbb{P}(A_j \mid S). \quad (8)$$

High unconditional correlation can arise even when (8) holds. By contrast, strategic dependence exists when one agent’s policy responds to the behavior, messages, beliefs, or anticipated actions of another. A useful intervention-based definition is therefore:

Definition 3.1 (Strategic dependence). *Agents i and j exhibit strategic dependence relative to probe set $\mathcal{Q}$ if there exists a controlled intervention $q \in \mathcal{Q}$ on communication, information, continuation, incentives, or counterpart behavior such that*

$$\mathcal{L}(A_i \mid q, S) \neq \mathcal{L}(A_i \mid q_0, S), \quad (9)$$

where q_0 is the baseline condition and the difference is attributable to the altered strategic channel rather than a change in the common signal S .

This definition captures the experimental principle of *Cooperation Beyond Correlation*: causal intervention, rather than action similarity alone, is required to distinguish strategic collaboration from shared-cause correlation (Reynoso, 2026a). For cybersecurity, the implication is immediate. Correlated actions by several agents are an incident indicator, not a causal diagnosis.

3.2 Voluntary information sharing and constructive collaboration

Collaboration can arise without central command when information is complementary. Let agent i possess private evidence e_i about an unknown state ω , and let $m_i \in \{0, 1\}$ indicate whether the evidence is shared. With communication cost c_i and downstream utility U_i , voluntary sharing is individually rational when

$$\mathbb{E}[U_i \mid m_i = 1] - c_i > \mathbb{E}[U_i \mid m_i = 0]. \quad (10)$$

The collective value of communication can exceed its individual value when evidence is complementary. *Constructive Collaboration* operationalizes this mechanism in a distributed scientific task, while retaining isolation and forced-sharing counterfactuals (Reynoso, 2026b). The cybersecurity relevance is not that constructive collaboration is dangerous; it is that the same decision architecture—local information, optional communication, and joint payoff effects—can sustain harmful cooperation when the joint objective is misaligned with institutional welfare.

3.3 Initial conditions and collaboration topology

Collaboration need not emerge uniformly. Let $\phi \in \Phi$ denote an initial-condition vector containing trust, reward alignment, communication cost, connectivity, and exploratory noise. A dynamic interaction rule

$$x_{t+1} = F(x_t; \phi) + \varepsilon_t \quad (11)$$

partitions the parameter space into basins such as

$$\Phi = \Phi_{\text{collab}} \cup \Phi_{\text{mixed}} \cup \Phi_{\text{fragmented}}. \quad (12)$$

The boundaries of these regions can be security-relevant because small changes in permissions, communication cost, memory, or reward alignment can move a system between qualitatively different collective regimes. *Parameterized Seeding and Topology* studies precisely this phase-topology logic in a controlled synthetic system (Reynoso, 2026c). A defender should therefore expect local sensitivity and regime shifts rather than a single global “cooperation parameter.”

3.4 Rationality calibration and representation robustness

Game theory is used here as a structural hypothesis about incentives, not as a metaphysical claim about machine cognition. A policy can be calibrated against equilibrium concepts by asking whether its choices are consistent with best responses or Nash equilibria in canonical games and whether that consistency transfers across equivalent representations. *Rationality Calibration in Canonical Games* uses canonical 2×2 games as calibration devices rather than as certification of universal rationality (Reynoso, 2026d).

This distinction is important because the defender’s model can be wrong in two different ways: the agent can deviate from the assumed solution concept, or the defender can fail to recognize an equivalent game because its representation changes. The later strategic-opacity analysis focuses on the second failure while allowing bounded rationality in the first.

3.5 Repeated games, memory, and reputation

In repeated interaction, a one-shot security analysis can be structurally misleading. Memory and reputation can transform defection-dominant stage incentives into cooperative long-run equilibria. In a repeated Prisoner’s Dilemma, a grim-trigger cooperation condition such as (2) illustrates how continuation value can sustain conduct that is not optimal in the stage game.

Repeated Games, Memory, Reputation, and an LLM Player treats memory interventions, public reputation, implementation noise, and an LLM-based decision rule as separate experimental layers (Reynoso, 2026e). For cybersecurity, repeated interaction explains why harmful collaboration can persist even when no single action reveals a binding commitment. The strategic object can live in history-dependent policies rather than in the current message alone.

3.6 Coalitions, internal value, and harmful externalities

A coalition game is defined by a characteristic function $v : 2^{\mathcal{N}} \rightarrow \mathbb{R}$ giving the value that each coalition can generate. An allocation $x \in \mathbb{R}^{|\mathcal{N}|}$ is in the core when

$$\sum_{i \in \mathcal{N}} x_i = v(\mathcal{N}) \quad \text{and} \quad \sum_{i \in S} x_i \geq v(S) \quad \forall S \subseteq \mathcal{N}. \quad (13)$$

Coalitions and Collaborative Equilibria shows why coalition efficiency, allocation, and stability are distinct objects (Reynoso, 2026f). This distinction matters for AI security because the institution can face a coalition that is internally stable even though its behavior creates a negative externality. The relevant security variable is therefore not “cooperation” in the abstract, but the relationship between coalition value and institutional harm.

The foundational studies imply a precise research stance. When autonomous agents collaborate in a way that becomes consequential, the defender should begin with the hypothesis that the behavior may be strategically rational under some hidden game. That hypothesis is neither an accusation of malicious intent nor a presumption of perfect optimization. It is a scientific prior that creates testable predictions about communication, incentives, repetition, reputation, coalition structure, and responses to intervention. The rest of the paper asks how that hidden game can be inferred and controlled when its representation is incomplete, potentially opaque, and possibly strategically chosen.

4 A Partially Observed Bayesian Stochastic Game for Cybersecurity

4.1 Environment

Let the cyber environment be a finite or measurable stochastic game

$$\mathcal{G}_\theta = \left\langle \mathcal{N}, \mathcal{X}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, P_\theta, \{u_i^\theta\}_{i \in \mathcal{N}}, \mathcal{O}, O_\theta \right\rangle, \quad (14)$$

where $\mathcal{N}$ is the set of strategic entities, $x_t \in \mathcal{X}$ is the latent system state, $a_{i,t} \in \mathcal{A}_i$ is player i ’s action, P_θ is the transition kernel, u_i^θ is player i ’s utility, $o_t \in \mathcal{O}$ is the defender’s observation, and $O_\theta(o_t | x_t, a_t)$ is the observation kernel. The latent parameter or mechanism $\theta \in \Theta$ can encode attacker type, coordination mechanism, reward dependence, persistence, communication dependence, or other structural features.

The defender need not be one of the strategic players in the underlying interaction. It can be an institutional controller that observes the system and chooses two types of actions: diagnostic probes $q_t \in \mathcal{Q}$ and security controls $c_t \in \mathcal{C}$. A probe changes the information structure or a bounded environmental feature in order to reveal mechanism dependence; a control is chosen primarily to reduce harm.

The defender maintains a belief

$$b_t(\theta) = \mathbb{P}(\theta \mid h_t), \quad (15)$$

where h_t is the history of representations, observations, probes, and outcomes. For finite Θ , Bayes' rule gives

$$b_{t+1}(\theta) = \frac{p(y_{t+1} \mid \theta, q_t, h_t) b_t(\theta)}{\sum_{\theta' \in \Theta} p(y_{t+1} \mid \theta', q_t, h_t) b_t(\theta')}. \quad (16)$$

4.2 Representation layer

The defender generally does not observe $\mathcal{G}_\theta$ directly. Instead it observes an artifact

$$z_t = R_{\gamma_t}(\mathcal{G}_\theta, x_t, h_t) + \varepsilon_t, \quad (17)$$

where R_γ is a representation map indexed by transformation parameter γ and ε_t captures noise, missingness, aggregation, or logging error. A central theme is that some transformations change only representation while others change the strategic object itself.

Definition 4.1 (Strategic equivalence). *Two finite normal-form games G and G' are strategically equivalent if there exist bijections over players and actions and, for each player i , constants $\alpha_i > 0$ and $\beta_i \in \mathbb{R}$ such that the transformed utilities satisfy*

$$u'_{\pi(i)}(\sigma(a)) = \alpha_i u_i(a) + \beta_i \quad (18)$$

for every action profile a , where π permutes players and σ denotes the induced action relabeling/permutation.

The definition can be generalized, but it deliberately uses positive affine transformations rather than arbitrary monotone transforms. Positive affine transformations preserve expected-utility comparisons and therefore preserve mixed as well as pure best responses.

Theorem 4.1 (Equilibrium invariance under admissible representation). *Let G' be obtained from a finite normal-form game G by player permutation, action bijections, and player-specific positive affine utility transformations. Then the best-response correspondences and Nash equilibrium set of G' are the images of those of G under the corresponding permutations.*

Proof. Fix player i and opponents' mixed strategy μ_{-i} . For any two actions a_i and a'_i , expected transformed utility satisfies

$$\mathbb{E}[u'_i(a_i, \mu_{-i})] - \mathbb{E}[u'_i(a'_i, \mu_{-i})] = \alpha_i (\mathbb{E}[u_i(a_i, \mu_{-i})] - \mathbb{E}[u_i(a'_i, \mu_{-i})]). \quad (19)$$

Because $\alpha_i > 0$, the sign of every expected-utility difference is unchanged. Hence the best-response set is preserved up to relabeling. A mixed profile is a Nash equilibrium if and only if each player's support lies in that player's best-response set. Therefore equilibrium is preserved under the induced bijections. $\square$

This theorem is elementary but foundational for *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment*: a difficult public artifact can be accepted only after an inverse certificate verifies that it belongs to the equivalence class of the source game. The security implication is subtle. A defender can be wrong about the strategic structure even when the structure itself has not changed.

5 Strategic Opacity and the Observer-Relative Game

5.1 Opacity as excess Bayes risk

A representation is not intrinsically opaque. It is opaque relative to an observer class, a model class, and a diagnostic budget. Let $D \in \mathfrak{D}$ be a decoder or inference procedure with computational budget B . Let $\ell_I(\hat{\theta}, \theta)$ be an identification loss. Define the optimal observer-relative Bayes risk under representation R_γ by

$$\mathcal{R}_{\mathfrak{D},B}(\gamma) = \inf_{D \in \mathfrak{D}(B)} \mathbb{E} [\ell_I(D(Z_\gamma), \Theta)]. \quad (20)$$

Definition 5.1 (Strategic opacity). *Relative to a canonical representation γ_0 , the strategic opacity of γ is*

$$\mathcal{O}_{\mathfrak{D},B}(\gamma; \gamma_0) = \mathcal{R}_{\mathfrak{D},B}(\gamma) - \mathcal{R}_{\mathfrak{D},B}(\gamma_0). \quad (21)$$

An information-theoretic analogue is

$$\mathcal{O}_H(\gamma; \gamma_0) = H(\Theta \mid Z_\gamma) - H(\Theta \mid Z_{\gamma_0}). \quad (22)$$

The Bayes-risk definition is especially useful because two observers can have different opacity even with the same data. A symbolic decoder, a frontier reasoning model, and a low-cost classifier may occupy different effective information sets.

Proposition 5.1 (Opacity does not imply strategic change). *There exist strategically equivalent games $G \sim G'$ and bounded observer classes $\mathfrak{D}(B)$ such that $\mathcal{O}_{\mathfrak{D},B}(G'; G) > 0$.*

Proof sketch. Choose an observer class whose internal representation or search depth is sufficient for a canonical matrix but insufficient for a sufficiently long invertible encoding that remains inside the admissible equivalence grammar. Because the mapping is invertible, an unbounded oracle can reconstruct G exactly; because the bounded class cannot execute the inverse map within budget B , its identification risk is larger. Hence strategic equivalence and observer-relative opacity coexist. $\square$

This is the formal version of *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment*'s empirical design. The canonical control estimates baseline recognizability. The encoded arm changes the representation while Python verifies inverse recovery. The treatment effect of encoding is therefore a causal effect of representation within the synthetic environment, not a confounding change in the source game.

5.2 Cybersecurity interpretation

Cyber telemetry is rarely an invariant representation. The same underlying dependency can be distributed across system logs, tool outputs, process names, service identities, network events,

retry sequences, and model-generated summaries. Normalization can help, but normalization can also collapse distinctions. Conversely, attackers or autonomous systems may change labels and surface organization while preserving the same strategic dependency. This motivates an operational concept of a *strategic signature*: not a literal indicator of compromise, but an invariant relationship among incentives, dependencies, responses to intervention, and equilibrium behavior.

The implication is not that security teams should ignore signatures. It is that semantic recognition and strategic recognition are different. A representation audit asks: which transformations of the observed data should leave the security conclusion invariant, and which transformations would imply a genuinely different game?

6 Active Inverse Games: Diagnosis as Experimental Design

6.1 Latent mechanism and probe model

Let $\theta \in \Theta$ denote the hidden mechanism. In *Inverse-Game Diagnosis under Scarce Experimental Budgets* a useful parameterization is

$$\vartheta = (b, \delta, c, \beta, \rho, \sigma, \kappa), \quad (23)$$

where b is a baseline event rate and the remaining coordinates represent sensitivity to future interaction, communication, payoff incentives, partner identity, shared signals, and coordination thresholds. The exact semantics can vary by domain. The important point is that the hidden object is structural and continuous, not merely a class label.

A probe $q \in \mathcal{Q}$ induces an observation distribution

$$Y \mid (\theta, q) \sim p_{\theta}^q(y). \quad (24)$$

Given current belief $b(\theta)$, the expected information gain from q is

$$\text{EIG}(q; b) = H(b) - \mathbb{E}_{Y \sim p(\cdot \mid q, b)} \left[H(b^{q, Y}) \right] = I(\Theta; Y \mid q, b), \quad (25)$$

where $b^{q, Y}$ is the posterior after observing Y .

Pure information gain is useful but not sufficient for cybersecurity. An experiment can be highly informative about a distinction that does not affect the optimal response. The control-relevant objective is instead

$$q^*(b) \in \arg \min_{q \in \mathcal{Q}} \left\{ c_q + \mathbb{E}_Y \left[\min_{a \in \mathcal{C}} \mathbb{E}_{\theta \sim b^{q, Y}} L(a, \theta) \right] \right\}, \quad (26)$$

where c_q includes direct experimental cost and, in fast-moving settings, the harm associated with diagnostic delay.

Equation (26) is the central extension from Bayesian experimental design to cyber intervention. It asks which experiment most improves the eventual control decision, not which experiment most reduces entropy in the abstract.

6.2 Observational equivalence and identifiability

Definition 6.1 (Probe-relative observational equivalence). *For a probe set $\mathcal{Q}_B \subseteq \mathcal{Q}$, mechanisms θ and θ' are observationally equivalent, written $\theta \sim_{\mathcal{Q}_B} \theta'$, if*

$$p_\theta^q(\cdot) = p_{\theta'}^q(\cdot) \quad \text{for every } q \in \mathcal{Q}_B. \quad (27)$$

No inference method can distinguish mechanisms in the same observational equivalence class using only those probes. Thus a weak empirical result can reflect underidentification rather than weak intelligence.

Theorem 6.1 (Finite-type posterior identification under active probing). *Let Θ be finite and let the true mechanism be θ^* . Suppose that for every $\theta \neq \theta^*$ there exists a probe q selected infinitely often such that*

$$D_{\text{KL}}(p_{\theta^*}^q \| p_\theta^q) > 0. \quad (28)$$

Assume conditionally independent observations given θ and the selected probes, with positive prior mass on θ^ . Then the posterior satisfies*

$$b_t(\theta^*) \rightarrow 1 \quad \text{almost surely.} \quad (29)$$

Proof sketch. For any false θ , the log posterior odds relative to θ^* equal prior log odds plus the accumulated log likelihood ratio. On every discriminating probe selected infinitely often, the strong law of large numbers implies that the average log likelihood ratio converges to $-D_{\text{KL}}(p_{\theta^*}^q \| p_\theta^q) < 0$. Hence the false-type posterior odds converge to zero. Because Θ is finite, posterior mass concentrates on θ^* . $\square$

The theorem explains why the probe policy matters. If an adaptive observer repeatedly chooses probes that fail to separate the remaining hypotheses, no amount of model intelligence can rescue identification. Conversely, a modest model can perform well if the experimental design supplies sharply discriminating evidence.

6.3 Scarce evidence budgets

Inverse-Game Diagnosis under Scarce Experimental Budgets deliberately constrains the observer. A large probe menu is paired with a finite evidence budget B and a small number of decision rounds. Let n_t be the number of samples allocated to probe q_t . The budget constraint is

$$\sum_{t=1}^{T_q} c(q_t, n_t) \leq B. \quad (30)$$

The observer jointly chooses q_t and n_t . Repetition is permitted because precision can be more valuable than breadth after a promising mechanism has been isolated.

The study's held-out counterfactual design is methodologically important. If the observer estimates a structural model $\hat{\theta}$, it must predict outcomes for interventions not used during diagnosis:

$$\hat{p}(q_{\text{hold}}) = p_{\hat{\theta}}^{q_{\text{hold}}}. \quad (31)$$

The primary score is therefore not merely classification accuracy but counterfactual prediction

error, such as

$$\text{MAE}_{\text{cf}} = \frac{1}{|\mathcal{Q}_H|} \sum_{q \in \mathcal{Q}_H} |\hat{p}(q) - p_{\theta^*}^q|. \quad (32)$$

This is a stronger test of mechanism recovery because labels can be correct for the wrong reason, while counterfactual generalization requires at least partial reconstruction of response structure.

7 Decision Equivalence and the Amount of Diagnosis Actually Needed

The cyber defender does not necessarily need to identify θ exactly. It needs enough information to choose a good control.

Definition 7.1 (Decision equivalence). *Mechanisms θ and θ' are decision equivalent under feasible control set $\mathcal{C}$ and loss L if their loss vectors differ only by an action-independent constant, or more weakly if they induce the same optimal action set:*

$$\arg \min_{a \in \mathcal{C}} L(a, \theta) = \arg \min_{a \in \mathcal{C}} L(a, \theta'). \quad (33)$$

We write $\theta \sim_L \theta'$.

Proposition 7.1 (No decision value from distinctions inside a loss-equivalence class). *If $L(a, \theta) = L(a, \theta') + K(\theta, \theta')$ for all $a \in \mathcal{C}$, then any information that distinguishes θ from θ' has zero value for control selection under L .*

Proof. The additive constant cancels from every pairwise comparison of actions. Hence the optimal action is identical under both mechanisms, and any posterior refinement that only redistributes mass between θ and θ' leaves expected action losses shifted by a common constant. $\square$

This proposition resolves an important evaluation problem. In *Inverse-Game Diagnosis under Scarce Experimental Budgets*, a model can fail a canonical label while reconstructing the intervention response accurately. In *Theory-Guided versus Model-Free Intervention*, a model can misclassify the source yet choose an almost optimal action. The policy-relevant state space is therefore the quotient $\Theta / \sim_L$, not necessarily the finest semantic taxonomy.

This suggests a hierarchy:

$$\boxed{\text{strategic equivalence} \rightarrow \text{observational equivalence} \rightarrow \text{decision equivalence.}} \quad (34)$$

Cybersecurity analysis should distinguish these layers explicitly. A representation problem belongs to the first quotient; an identification problem to the second; an intervention problem to the third.

8 Time-Bounded Intervention and the Value of Theory

8.1 Cyber loss function

Let $h_\theta(t) \geq 0$ be the uncontrolled harm rate under mechanism θ . A control a has implementation time $T_I(a)$, effectiveness $e(a, \theta) \in [0, 1]$, and direct control cost $k(a)$. Suppose diagnosis takes

T_D . Define activation time

$$\tau(a) = T_D + T_I(a). \quad (35)$$

Let H_θ denote an irreversible-harm exposure threshold and $\lambda > 0$ the penalty multiplier. Over horizon T , define

$$\begin{aligned} L(a, \theta; T_D) = & \int_0^{\tau(a)} h_\theta(t) dt + \int_{\tau(a)}^T (1 - e(a, \theta)) h_\theta(t) dt + k(a) \\ & + \lambda \left[\int_0^{\tau(a)} h_\theta(t) dt - H_\theta \right]_+. \end{aligned} \quad (36)$$

Equation (36) generalizes *Theory-Guided versus Model-Free Intervention*'s synthetic loss oracle. It makes the cost of analysis explicit: deeper diagnosis does not enter the loss only as compute expenditure; it delays the point at which the targeted control becomes effective.

8.2 Model-free, theory-guided, and oracle policies

A robust model-free policy can be represented by

$$a_{\text{MF}} \in \arg \min_{a \in \mathcal{C}} \sup_{\theta \in \Theta} L(a, \theta; 0), \quad (37)$$

or by a risk measure such as CVaR_α . A theory-guided Bayesian policy with posterior b_D after diagnosis chooses

$$a_{\text{TG}}(b_D) \in \arg \min_{a \in \mathcal{C}} \mathbb{E}_{\theta \sim b_D} [L(a, \theta; T_D)]. \quad (38)$$

The private oracle is

$$a_{\text{OR}}(\theta) \in \arg \min_{a \in \mathcal{C}} L(a, \theta; 0). \quad (39)$$

The oracle is not a deployable policy; it defines a lower-loss benchmark and hence decision regret.

8.3 Static value of information

With zero diagnostic cost and zero delay, information has weakly nonnegative value in a Bayesian decision problem.

Proposition 8.1 (Static value of information). *Let S be any signal about Θ , and suppose acquisition of S is costless and does not delay action. Then*

$$\mathbb{E} \left[\min_a \mathbb{E}[L(a, \Theta; 0) \mid S] \right] \leq \min_a \mathbb{E}[L(a, \Theta; 0)]. \quad (40)$$

Proof. After observing S , the decision maker can always ignore it and choose the prior-optimal action. Therefore the set of feasible contingent decision rules contains the constant prior-optimal rule, so the expected optimum cannot be worse. $\square$

The proposition is a decision-theoretic version of the intuition behind Blackwell informativeness. Cybersecurity differs because signals are not free in time.

8.4 When delay reverses the value of theory

Define the *static targeting gain*

$$G_{\text{info}} = \mathbb{E}[L(a_{\text{MF}}, \Theta; 0)] - \mathbb{E}[L(a_{\text{TG}}(b_D), \Theta; 0)], \quad (41)$$

where the second term evaluates the theory-guided action as if it could be implemented instantly. Define the expected delay burden

$$C_{\text{delay}}(T_D) = \mathbb{E}[L(a_{\text{TG}}(b_D), \Theta; T_D) - L(a_{\text{TG}}(b_D), \Theta; 0)] + C_D, \quad (42)$$

where C_D is any direct diagnostic cost.

Theorem 8.1 (Value-of-theory threshold). *Theory-guided diagnosis reduces expected loss relative to the model-free benchmark if and only if*

$$G_{\text{info}} > C_{\text{delay}}(T_D). \quad (43)$$

If $C_{\text{delay}}(T_D)$ is continuous and strictly increasing in T_D , there is at most one threshold T_D^ satisfying equality, and theory is beneficial only for $T_D < T_D^*$ whenever such a threshold exists.*

Proof. Subtract expected theory-guided loss with delay from expected model-free loss and add/subtract the theory-guided zero-delay loss. The difference is exactly $G_{\text{info}} - C_{\text{delay}}(T_D)$. The monotonicity statement follows immediately from strict increase of the delay burden. $\square$

The theorem formalizes *Theory-Guided versus Model-Free Intervention*'s central idea. More accurate diagnosis can worsen the decision if the additional precision is purchased with too much exposure time.

8.5 A closed-form illustration

Suppose harm is constant, $h_\theta(t) = h_\theta$, the selected target control has effectiveness e_θ , and no irreversibility penalty binds. Then

$$L(a, \theta; T_D) = h_\theta [T_D + T_I(a)] + h_\theta(1 - e_\theta) [T - T_D - T_I(a)] + k(a). \quad (44)$$

Differentiating with respect to T_D gives

$$\frac{\partial L}{\partial T_D} = h_\theta e_\theta. \quad (45)$$

The marginal cost of diagnostic delay is therefore larger when the eventual targeted control is highly effective. Paradoxically, the more useful the control, the more costly it is to postpone it. This is one reason a containment bridge can be valuable.

9 Contain–Diagnose–Target as an Option-Preserving Security Architecture

Let a_0 be an immediate containment action with implementation time τ_0 , effectiveness $e_0(\theta)$, and cost k_0 . Let a_1 be the diagnosis-contingent targeted action activated at $T_D + T_I(a_1)$. After

a_1 activates, suppose combined effectiveness is

$$e_{01}(\theta) = 1 - (1 - e_0(\theta))(1 - e_1(\theta)), \quad (46)$$

possibly capped below one to reflect residual risk.

The hybrid loss is piecewise:

$$\begin{aligned} L_H = & \int_0^{\tau_0} h_\theta(t) dt + \int_{\tau_0}^{\tau_1} (1 - e_0(\theta)) h_\theta(t) dt \\ & + \int_{\tau_1}^T (1 - e_{01}(\theta)) h_\theta(t) dt + k_0 + k_1 + \Phi_H, \end{aligned} \quad (47)$$

where $\tau_1 = T_D + T_I(a_1)$ and Φ_H is any irreversibility penalty.

Theorem 9.1 (Sufficient condition for hybrid dominance over diagnosis-first). *Assume constant harm rate $h > 0$, $0 < \tau_0 < \tau_1 < T$, no difference in the ultimate targeted action, and no additional irreversibility penalty under the hybrid. The hybrid policy has lower loss than diagnosis-first targeting whenever*

$$h e_0(\tau_1 - \tau_0) + h(e_{01} - e_1)(T - \tau_1) > k_0. \quad (48)$$

Proof. Subtract hybrid loss from diagnosis-first loss. Before τ_0 the policies coincide. Between τ_0 and τ_1 , containment reduces harm by $h e_0$ per unit time. After τ_1 , any incremental combined-effectiveness gain contributes the second term. The hybrid pays the additional containment cost k_0 . The hybrid dominates exactly when avoided harm exceeds that cost. $\square$

If the diagnosis-first policy would cross an irreversible threshold while containment keeps cumulative exposure below it, the hybrid can dominate by a much larger margin because it also avoids the penalty term. This is the economic meaning of containment as an option-preserving action: it buys time for information acquisition while keeping the system inside a viable region.

A more general safe-control formulation defines a safe set $\mathcal{S} \subseteq \mathcal{X}$ and requires

$$\mathbb{P}(x_t \in \mathcal{S} \forall t \leq T) \geq 1 - \epsilon. \quad (49)$$

The probe-selection problem then becomes a constrained POMDP: gather information only through experiments that preserve the probability of remaining inside the safety envelope.

10 Provenance as Stage-Dependent Information Capital

The time-bounded intervention problem in *Theory-Guided versus Model-Free Intervention* is an incident-level stopping problem: should the defender continue diagnosing or intervene now? *Provenance, Value of Information, and Incident Response* extends the horizon. Once a system is contained, the institution faces a sequence of distinct decisions whose information requirements need not coincide. Prevention asks how architecture should change before recurrence; containment asks how to arrest current harm; remediation asks which causal dependency must be repaired; redeployment asks what autonomy and permissions can be restored. The same posterior can therefore be sufficient at one stage and insufficient at another.

10.1 Trace provenance and causal provenance

Let $\mathcal{P}_t$ denote the trace-provenance record available at time t . Abstractly, $\mathcal{P}_t$ can be represented as a labeled directed graph

$$\mathcal{P}_t = (V_t, E_t, \lambda_t), \quad (50)$$

whose nodes encode relevant entities, activities, agents, model versions, tools, messages, state transitions, and observations, and whose edges encode use, generation, derivation, attribution, and temporal relations. This abstraction is deliberately compatible with the entity–activity–agent logic of W3C PROV without requiring a particular storage technology (Moreau and Missier, 2013).

Trace provenance should be distinguished from the latent *causal provenance* $\theta \in \Theta$ that generates the incident. The former is evidence; the latter is a strategic object of inference. In *Provenance, Value of Information, and Incident Response* the latent vector is stylized as

$$\theta = (\theta_{\text{cm}}, \theta_{\text{dc}}, \theta_{\text{ri}}, \theta_{\text{eo}}, \theta_{\text{so}}), \quad (51)$$

representing common-mode dependence, direct communication dependence, reward/incentive dependence, external orchestration dependence, and strategic opacity. The final component links provenance directly to *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment*: difficult recognition can itself be causal evidence that a familiar strategic structure is present under a transformed representation.

Let the incident lifecycle stage be

$$s \in \mathcal{S} = \{\text{P, C, R, D}\}, \quad (52)$$

for prevention, containment, remediation, and redeployment, with stage-specific action set $\mathcal{A}_s$ and loss $L_s(a, \theta; \delta)$, where δ is the delay incurred before the action becomes effective.

10.2 Stage-specific Bayes risk

Definition 10.1 (Stage-specific Bayes risk). *For belief $b \in \Delta(\Theta)$, stage s , information delay δ , and information cost c , define*

$$R_s(b; \delta, c) = \min_{a \in \mathcal{A}_s} \mathbb{E}_{\theta \sim b} [L_s(a, \theta; \delta)] + c. \quad (53)$$

Equation (53) is the fundamental object for provenance economics. Information is not valued by the entropy it removes but by the reduction in stage-specific Bayes risk it enables. The same posterior b can therefore imply different information-acquisition decisions at different stages because both $\mathcal{A}_s$ and L_s vary with s .

The expected value of perfect information at stage s is

$$\text{EVPI}_s(b) = R_s(b; 0, 0) - \mathbb{E}_{\theta \sim b} \left[\min_{a \in \mathcal{A}_s} L_s(a, \theta; 0) \right]. \quad (54)$$

For an additional provenance experiment q with outcome Y , direct cost c_q , and delay Δt_q , define the net expected value of sample information as

$$\text{NetEVSI}_s(q; b) = R_s(b; 0, 0) - \mathbb{E}_Y \left[R_s(b^{q,Y}; \Delta t_q, c_q) \right]. \quad (55)$$

An institution should acquire the evidence only when (55) is positive. *Provenance, Value of Information, and Incident Response* implements this comparison between fast and deep provenance packets and makes the delay component stage-specific.

Proposition 10.1 (Zero value of perfect provenance under a common Bayes action). *Fix a stage s and posterior b . If there exists $a_s^* \in \mathcal{A}_s$ such that*

$$a_s^* \in \arg \min_{a \in \mathcal{A}_s} L_s(a, \theta; 0) \quad \text{for every } \theta \in \text{supp}(b), \quad (56)$$

then

$$\text{EVPI}_s(b) = 0. \quad (57)$$

Proof. Under (56), the same action minimizes loss state by state. Hence

$$R_s(b; 0, 0) = \mathbb{E}_b[L_s(a_s^*, \Theta; 0)] \quad (58)$$

$$= \mathbb{E}_b \left[\min_{a \in \mathcal{A}_s} L_s(a, \Theta; 0) \right], \quad (59)$$

which makes the difference in (54) equal to zero. $\square$

Proposition 10.1 formalizes a central *Provenance, Value of Information, and Incident Response* insight. A signal can be highly informative in the Shannon sense and still have no institutional value for the current decision. If containment is robust across all high-probability mechanisms, attribution can wait even when the posterior remains diffuse.

Corollary 10.1 (Entropy reduction need not imply decision value). *Suppose a costless experiment strictly reduces expected posterior entropy at stage s , but condition (56) holds both before and after the experiment. Then the experiment has zero gross decision value. If the experiment has strictly positive cost or delay burden, its net value is negative.*

This corollary separates information quantity from information value in exactly the sense emphasized by the classical value-of-information literature (Howard, 1966). It also explains why a cybersecurity organization can rationally stop investigating before attribution is complete.

10.3 Stage-specific decision equivalence

Definition 10.2 (Stage-specific decision equivalence). *Two mechanisms θ_i and θ_j are decision-equivalent at stage s , written*

$$\theta_i \sim_{L,s} \theta_j, \quad (60)$$

if their sets of optimal stage actions coincide:

$$\arg \min_{a \in \mathcal{A}_s} L_s(a, \theta_i; 0) = \arg \min_{a \in \mathcal{A}_s} L_s(a, \theta_j; 0). \quad (61)$$

The quotient $\Theta / \sim_{L,s}$ is stage dependent. In particular, broad containment controls can create large equivalence classes, while targeted remediation can induce a much finer partition. This provides a formal interpretation of the institutional rule

resolve uncertainty only to the level required by the next consequential decision.

(62)

Let $C_s(\theta)$ denote the equivalence class of θ at stage s . A posterior is $(1 - \alpha)$ *stage sufficient* if some equivalence class contains at least $1 - \alpha$ posterior mass. This criterion can be more operationally meaningful than a global classification-confidence threshold because it asks whether residual uncertainty can still change the action.

10.4 Preserved provenance as option value

Evidence that is not useful for containment can still be valuable later. Let $\mathcal{P}$ denote an existing provenance archive, and let $\tilde{\mathcal{P}}$ be an enriched archive preserved at incremental storage and integrity cost c_P . Define the gross continuation value of preserved provenance at future stage s' by

$$V_{P,s'} = V_{s'}(b, \mathcal{P}) - V_{s'}(b, \tilde{\mathcal{P}}), \quad (63)$$

where lower V denotes lower minimum expected loss.

Proposition 10.2 (Weakly nonnegative gross option value of ignorable provenance). *If preserving $\tilde{\mathcal{P}}$ does not constrain future actions and the institution can ignore the additional evidence, then its gross option value is weakly nonnegative.*

Proof. The decision rule available without the enriched archive remains feasible when the archive is available. Therefore the minimum expected future loss with $\tilde{\mathcal{P}}$ cannot exceed the minimum expected future loss without it. $\square$

This is a Blackwell-style argument applied to institutional evidence. It does not imply that all logging is socially optimal. Once storage cost, latency, privacy, confidentiality, integrity, and operational complexity are included, the *net* value of preservation can be negative. The proposition says only that optional trustworthy evidence has nonnegative gross value because the institution can choose not to use it.

10.5 Redeployment as a provenance-constrained autonomy decision

The redeployment stage can be represented more generally by an autonomy level $u \in [0, 1]$, where $u = 0$ corresponds to quarantine or shadow operation and $u = 1$ to full autonomy. Let $\ell(u, \theta)$ be residual redeployment loss, increasing in autonomy for high-risk mechanisms, and let $\bar{r}$ be the institution's risk budget. A provenance-aware autonomy envelope is

$$u^*(b) = \sup \{u \in [0, 1] : \text{CVaR}_\alpha(\ell(u, \Theta) \mid b) \leq \bar{r}\}. \quad (64)$$

The discrete *Provenance*, *Value of Information*, and *Incident Response* actions *full*, *bounded*, *quarantine*, and *shadow mode* can be understood as coarse points on this continuum. Provenance is valuable at redeployment precisely when it changes the admissible autonomy envelope.

10.6 Provenance acquisition as a stage-specific stopping problem

At stage s , the institution should continue provenance acquisition only while

$$\max_{q \in \mathcal{Q}_s} \text{NetEVSI}_s(q; b) > 0. \quad (65)$$

Unlike a global attribution objective, (65) can prescribe immediate action at containment, deeper reconstruction at remediation, and still deeper validation before redeployment. Provenance is

therefore institutional capital with a maturity structure: its return depends on when it becomes usable and which decision it supports.

11 A Unified Belief-State Control Problem

The four cybersecurity layers can be combined in a single dynamic program. Let the defender's state be

$$s_t = (b_t, x_t, B_t, \tau_t, z_t, \sigma_t, \mathcal{P}_t), \quad (66)$$

where b_t is belief over mechanisms, x_t is the physical or cyber state, B_t is remaining diagnostic budget, τ_t is remaining time to a critical threshold, z_t is the current representation, $\sigma_t \in \mathcal{S}$ is the incident-response stage, and $\mathcal{P}_t$ is the preserved provenance record. At each step the defender can choose a diagnostic probe q , a containment action a_0 , a target control a , a provenance-preservation action, a stage transition, or a terminal decision.

A schematic Bellman equation is

$$V(s_t) = \min \left\{ \min_{a \in \mathcal{C}} [\ell(s_t, a) + \gamma \mathbb{E}V(s_{t+1})], \right. \\ \left. \min_{q \in \mathcal{Q}(B_t)} [c(q) + \ell_{\text{delay}}(s_t, q) + \gamma \mathbb{E}_{Y_q} V(s_{t+1}^{q,Y})] \right\}. \quad (67)$$

The defender therefore chooses between learning and acting. In contrast with standard active learning, the value of the next observation depends on how much harm accumulates while waiting and whether the probe itself perturbs the system.

To make the *Provenance, Value of Information, and Incident Response* extension explicit, let $p \in \mathcal{U}(\mathcal{P}_t)$ denote a provenance-preservation or integrity action with direct cost $c_P(p)$ and next archive $\mathcal{P}_{t+1}^p$. The Bellman operator can be augmented by a third branch,

$$\min_{p \in \mathcal{U}(\mathcal{P}_t)} [c_P(p) + \gamma \mathbb{E}V(b_t, x_t, B_t, \tau_t, z_t, \sigma_t, \mathcal{P}_{t+1}^p)], \quad (68)$$

and by stage-transition operators that replace L_s and $\mathcal{A}_s$ when the institution moves from containment to remediation or from validation to redeployment. The relevant sufficient statistic is therefore not merely a posterior over cyber mechanisms. It is a posterior together with the evidence archive and the stage-dependent action geometry.

Equation (67) clarifies the relationship among the computational studies. *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment* concerns the mapping from z_t to beliefs b_t when representation changes but the game does not. *Inverse-Game Diagnosis under Scarce Experimental Budgets* concerns the information branch: selecting q to improve b_t . *Theory-Guided versus Model-Free Intervention* concerns the switching condition between learning and immediate control. *Provenance, Value of Information, and Incident Response* adds σ_t and $\mathcal{P}_t$: the optimal amount of further attribution depends on the incident stage, and preserved evidence can create option value for remediation and redeployment even when it has little immediate containment value.

12 Cyber Mechanism Taxonomy and Safe Diagnostic Logic

The synthetic mechanism dimensions used across the computational studies have natural, high-level cybersecurity interpretations. Table 1 is deliberately non-operational: it maps causal structures to defensive diagnostic questions and containment classes rather than attack instructions.

For agentic AI, these mechanisms are especially relevant because the system boundary includes memory, tools, permissions, orchestration, and other agents. MITRE ATLAS and OWASP’s agentic-security work emphasize that AI systems must be analyzed as systems with broader attack and execution surfaces, not as isolated model endpoints (MITRE, 2026; OWASP, 2025). The game-theoretic contribution is to ask which strategic dependency explains observed behavior and which controlled perturbation would distinguish competing explanations.

13 Representation, Diagnosis, Intervention, and Provenance as Four Quotient Problems

The paper’s central mathematical synthesis can be stated in terms of quotient spaces.

Let $\mathfrak{G}$ be the space of games. Let Γ be a group or semigroup of admissible representation transformations. The strategic object is an equivalence class

$$[G]_{\Gamma} = \{R_{\gamma}(G) : \gamma \in \Gamma\}. \quad (69)$$

The defender’s first problem is to infer $[G]_{\Gamma}$ from the artifact rather than confuse one representative with a different game.

Next let Θ be the mechanism space and $\mathcal{Q}_B$ the probes feasible under budget B . Diagnostic data identify only the quotient

$$\Theta / \sim_{\mathcal{Q}_B}, \quad (70)$$

where mechanisms are grouped when they generate identical observation laws under all feasible probes.

Finally, the control problem may require only

$$\Theta / \sim_L, \quad (71)$$

where mechanisms are grouped when they induce the same relevant loss vector or optimal action. The security problem is therefore not to recover the finest possible hidden truth in every case. It is to refine the uncertainty set only until the remaining ambiguity is decision-irrelevant.

Provenance, Value of Information, and Incident Response adds a fourth quotient that is indexed by institutional stage. Let $\sim_{L,s}$ be the stage-specific decision-equivalence relation in (61). Then the amount of provenance required at stage s is determined by

$$\Theta / \sim_{L,s}. \quad (72)$$

The partition can be coarse at containment and fine at remediation. Consequently, there is no single globally sufficient attribution state. A belief that is adequate to contain safely can be inadequate to select a root-cause remediation or a redeployment autonomy level. The defender’s epistemic target therefore changes endogenously over the incident lifecycle.

Table 1: Strategic mechanism classes and defensive interpretation

Mechanism	Cyber interpretation	Safe diagnostic question	Candidate defensive class
Common cause / shared signal	Multiple agents respond to the same exogenous input, shared data source, shared policy state, or correlated environmental signal.	Does coordinated behavior weaken when the shared signal is removed, decorrelated, or replaced in a sandbox?	Signal suppression, input diversification, source isolation, independent validation.
Communication-mediated coordination	Joint behavior depends materially on direct or indirect agent-to-agent communication.	Does coordination collapse or change when communications are disabled or degraded in a controlled environment?	Communication isolation, permission scoping, message mediation.
Repeated incentive / persistence	Cooperation or harmful behavior is sustained by future interaction, memory, persistence, reputation, or repeated access.	Does behavior change when persistence or continuation is removed in a test environment?	Session isolation, credential rotation, memory/persistence controls, ephemeral execution.
Reward or objective misdesign	Behavior is generated by an optimization objective that makes the harmful state locally attractive.	Does bounded reward/objective perturbation materially change the behavior?	Reward redesign, policy constraints, verifier insertion, objective decomposition.
External orchestration	Multiple components depend on an external controller, integration, tool, workflow engine, or common command source.	Does behavior change when the external dependency is isolated in a governed sandbox?	External-channel isolation, allowlisting, brokered execution, dependency quarantine.
Opaque familiar game	A familiar strategic structure is present but transformed representation, decomposition, or telemetry makes it hard for a bounded observer to recover.	Can a representation-invariant reconstruction recover the same best-response or equilibrium structure?	Increase legibility, preserve representation lineage, reconstruct and validate the strategic mechanism, constrain redeployment until recovered.

This gives a formal stopping principle for diagnosis.

Definition 13.1 (Decision-sufficient posterior). *A posterior b is decision sufficient at tolerance ϵ if there exists an action a_b such that*

$$\mathbb{E}_{\theta \sim b} L(a_b, \theta) - \mathbb{E}_{\theta \sim b} \min_a L(a, \theta) \leq \epsilon. \quad (73)$$

A rational diagnostic policy should stop when the expected value of further belief refinement is smaller than its cost and delay. In this sense, “uncertainty” is not itself a failure. Residual uncertainty is acceptable if it lies within a decision-equivalent region.

14 Computational Research Program: From Cooperation to Provenance-Aware Control

The *Game Theory in the Age of AI* research program contains ten named computational studies whose logic is cumulative (Reynoso, 2026k). The first six establish the game-theoretic context for collaboration. The final four are the cybersecurity laboratories formalized most directly in this paper. Referring to the studies by title rather than by file order is important: each study represents a distinct scientific question and can ultimately be cited independently once the collection is published in the public repository.

14.1 Foundational studies of collaboration

Cooperation Beyond Correlation. This study asks whether controlled interventions can distinguish strategic cooperation from common-cause correlation. Its role in the present paper is foundational: correlated agent behavior is treated as an observation that requires causal discrimination, not as proof of collaboration. The cyber analogue is the difference between agents responding independently to the same signal and agents whose policies are strategically interdependent (Reynoso, 2026a).

Constructive Collaboration. This study models agents with complementary evidence that may voluntarily share information at a cost. It establishes a mechanism through which collaboration can be individually and collectively rational without being centrally commanded. The security implication is that collaboration can be an equilibrium response to incentives and information structure rather than an exceptional failure state (Reynoso, 2026b).

Parameterized Seeding and Topology. This study maps collaboration across controlled initial conditions and studies boundary sensitivity. Its contribution to the security framework is the idea that cooperative or coordinated regimes can occupy structured regions of parameter space. Small changes in trust, communication cost, reward alignment, connectivity, or exploration can move the system across phase boundaries (Reynoso, 2026c).

Rationality Calibration in Canonical Games. This study calibrates decision procedures against canonical game-theoretic solution concepts and representation controls. It motivates the use of game theory in this paper as a falsifiable structural benchmark rather than an assumption of perfect cognition. It also foreshadows the later distinction between strategic structure and surface representation (Reynoso, 2026d).

Repeated Games, Memory, Reputation, and an LLM Player. This study examines repeated interaction, memory, reputation, implementation noise, and an LLM player in the Prisoner’s Dilemma. It supplies the dynamic logic through which cooperation can persist even when one-shot incentives would predict otherwise, and it highlights how history-dependent policies complicate cyber diagnosis (Reynoso, 2026e).

Coalitions and Collaborative Equilibria. This study separates coalition efficiency, allocation, and stability in a transferable-utility setting. It motivates the paper’s distinction between agent welfare and institutional welfare: a coalition can be internally productive and stable while imposing a harmful externality on the surrounding organization (Reynoso, 2026f).

14.2 *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment: representation as the treatment*

Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment uses four canonical 2×2 games and an admissible encoding grammar (Reynoso, 2026g). The source game remains private. The representation designer can relabel players and actions, permute ordering, exchange roles, apply player-specific positive affine transformations, decompose transformed payoffs into accounting components, reorder outcome records, and add neutral contextual distractors. A deterministic Python oracle applies the recipe and verifies inverse reconstruction before the artifact reaches the observer. A canonical-control arm presents the same source game directly. The estimand is the change in identification behavior induced by representation while strategic structure is held fixed.

Let Y_{grc} be an identification outcome for game g , replication r , and condition $c \in \{0, 1\}$, with $c = 1$ denoting encoded representation. A basic treatment effect is

$$\Delta_R = \mathbb{E}[Y \mid c = 1] - \mathbb{E}[Y \mid c = 0]. \quad (74)$$

For binary accuracy, a hierarchical logistic model can account for game-family heterogeneity:

$$\text{logit } \mathbb{P}(Y_{grc} = 1) = \alpha_g + \beta_{Rc} + u_r. \quad (75)$$

14.3 *Inverse-Game Diagnosis under Scarce Experimental Budgets: active inverse diagnosis*

Inverse-Game Diagnosis under Scarce Experimental Budgets expands the probe space, imposes a scarce evidence budget, and hides a structural vector rather than merely a label (Reynoso, 2026h). The observer chooses both intervention and sample allocation, while passive and expected-information-gain comparators provide reference policies. Three probes are held out from diagnosis and used for counterfactual evaluation. The key estimands include closed-set calibration, latent-parameter error, and held-out counterfactual error.

Let M_{ir}^A and M_{ir}^P be held-out prediction MAE for adaptive and passive conditions on case i , replication r . The paired adaptive advantage is

$$\Delta_A = \mathbb{E} \left[M^P - M^A \right]. \quad (76)$$

Positive Δ_A means adaptive diagnosis improves generalization. A paired permutation test or hierarchical model should be preferred to an unpaired comparison because the same hidden

mechanism is reused across arms.

14.4 *Theory-Guided versus Model-Free Intervention: the realized value of theory*

Theory-Guided versus Model-Free Intervention defines source mechanisms through latent causal sensitivities such as communication dependence, future/persistence dependence, reward sensitivity, shared-signal dependence, and external orchestration (Reynoso, 2026i). The action space includes observation, rate limitation, communication isolation, persistence disruption, reward redesign, shared-signal suppression, external-channel isolation, and shutdown. GPT-4o mini is evaluated under model-free, fast theory-guided, and deep theory-guided information regimes; a hybrid condition combines immediate rate limitation with deep diagnosis. Minimax and private-oracle benchmarks bound performance.

The primary estimand is not diagnosis accuracy but loss reduction:

$$\Delta_T = \mathbb{E}[L_{\text{MF}} - L_{\text{TG}}]. \quad (77)$$

The paired hybrid advantage is

$$\Delta_H = \mathbb{E}[L_{\text{MF}} - L_{\text{Hybrid}}]. \quad (78)$$

Regret relative to the private oracle is

$$\text{Regret}_\pi(\theta) = L(a_\pi, \theta) - \min_a L(a, \theta). \quad (79)$$

14.5 *Provenance, Value of Information, and Incident Response: provenance as stage-dependent information capital*

Provenance, Value of Information, and Incident Response retains five provenance hypotheses—common mode, direct communication, reward/incentive, external orchestration, and opaque familiar game—and embeds them in a continuous latent provenance vector (Reynoso, 2026j). A private GPT-5.2 designer creates ambiguous incidents; deterministic Python constrains the hidden mechanism and generates nested fast and deep evidence packets; GPT-4o mini chooses stage-specific policies under provenance-blind, fast-provenance, and deep-provenance regimes. The stages are prevention, containment, remediation, and redeployment. A transparent Bayesian benchmark computes posterior entropy, EVPI, net EVSI of deeper provenance, and stage-specific decision-equivalence classes.

Let L_{is}^F and L_{is}^D denote realized loss at stage s under fast and deep provenance for incident i . Define the realized value of deeper provenance

$$\Delta_{P,s} = \mathbb{E}[L_{is}^F - L_{is}^D]. \quad (80)$$

This effect can be negative even when deep provenance improves classification, because deeper attribution has information and delay costs. A second estimand is the action-change rate

$$\chi_s = \mathbb{P}(a_{is}^F \neq a_{is}^D), \quad (81)$$

which measures whether the additional evidence is institutionally consequential. The normative benchmark is the sign and magnitude of NetEVSI_s in (55).

Table 2: Cumulative logic of the *Game Theory in the Age of AI* computational studies

Study	Core scientific question	Role in the cybersecurity framework
<i>Cooperation Beyond Correlation</i>	Can intervention distinguish strategic cooperation from common-cause correlation?	Establishes causal identification before attribution of strategic interdependence.
<i>Constructive Collaboration</i>	Can voluntary information sharing improve a distributed task?	Shows that uncommanded collaboration can be instrumentally rational under complementary information.
<i>Parameterized Seeding and Topology</i>	Does collaboration occupy structured regions of initial-condition space?	Introduces phase topology, regime boundaries, and sensitivity to system configuration.
<i>Rationality Calibration in Canonical Games</i>	Are decision policies consistent with canonical equilibrium concepts across representations?	Calibrates game-theoretic rationality and representation robustness.
<i>Repeated Games, Memory, Reputation, and an LLM Player</i>	How do repetition, memory, reputation, and noise affect cooperation?	Supplies dynamic mechanisms for persistence and history-dependent strategic behavior.
<i>Coalitions and Collaborative Equilibria</i>	When is a coalition efficient, stable, and acceptably rewarded?	Separates coalition value, stability, allocation, and harmful externalities.
<i>Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment</i>	Can an unchanged game become harder to recognize through representation alone?	Formalizes strategic opacity and the gap between objective and observer-relative games.
<i>Inverse-Game Diagnosis under Scarce Experimental Budgets</i>	Which scarce experiments best recover a hidden mechanism and predict unseen interventions?	Treats incident diagnosis as active inverse-game inference rather than passive classification.
<i>Theory-Guided versus Model-Free Intervention</i>	Does strategic diagnosis improve realized control after delay and cost?	Introduces time-bounded solvency, value of theory, and hybrid containment-diagnose-target policy.
<i>Provenance, Value of Information, and Incident Response</i>	When is deeper provenance worth acquiring across the incident lifecycle?	Formalizes stage-specific Bayes risk, EVPI/EVSI, decision equivalence, and autonomy-constrained redeployment.

15 Econometric and Statistical Design for Journal-Level Evaluation

The computational program becomes publishable only when model outputs are treated as data generated by a stochastic experimental process rather than as anecdotes. API-based models can be nondeterministic, hidden cases vary in difficulty, and game families induce heterogeneous treatment effects. A journal-level evaluation should therefore pre-specify estimands, replication, randomization, and uncertainty quantification.

15.1 Representation experiment

For *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment*, the cleanest design is paired within source game and replication. Each accepted encoded artifact has a canonical counterpart generated from the same source. Let A_{igr}^E and A_{igr}^C denote accuracy under encoded and canonical conditions. The paired difference

$$d_{igr} = A_{igr}^E - A_{igr}^C \quad (82)$$

can be aggregated with game fixed effects or a hierarchical model. Confidence and abstention should be analyzed separately because opacity can manifest as calibrated caution rather than error.

Calibration can be summarized by Brier score

$$BS = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K (p_{ik} - y_{ik})^2, \quad (83)$$

and log loss

$$LL = -\frac{1}{N} \sum_{i=1}^N \log p_{i,y_i}. \quad (84)$$

A representation that leaves accuracy unchanged but substantially increases entropy is still empirically interesting because it has reduced decisiveness.

15.2 Adaptive diagnosis experiment

For *Inverse-Game Diagnosis under Scarce Experimental Budgets*, repeated hidden cases should be generated across source families and decoy difficulty levels. The observer's probe sequence is endogenous. This is a feature, not a nuisance: sequence choice is itself an outcome. Useful statistics include probe-selection frequency, sample allocation, posterior entropy path, overlap with the EIG benchmark, and held-out counterfactual error.

A hierarchical specification for held-out error can be written

$$M_{irc} = \alpha_i + \delta_r + \beta_A \mathbf{1}\{c = A\} + \beta_B B_{ir} + \epsilon_{irc}, \quad (85)$$

where α_i absorbs hidden-case difficulty, δ_r replication effects, and B_{ir} may measure ambiguity or diagnostic separation. The coefficient β_A estimates whether adaptivity improves generalization conditional on case difficulty.

15.3 Intervention experiment

Theory-Guided versus Model-Free Intervention produces a continuous loss outcome. A paired loss regression can estimate regime effects:

$$L_{ic} = \alpha_i + \beta_F \mathbf{1}\{c = \text{fast}\} + \beta_D \mathbf{1}\{c = \text{deep}\} + \beta_H \mathbf{1}\{c = \text{hybrid}\} + \epsilon_{ic}. \quad (86)$$

Because the private oracle provides a case-specific lower benchmark, regret is an especially interpretable dependent variable. Tail risk should also be reported. A policy with slightly lower mean loss but much higher catastrophic tail loss may be unacceptable. For level α ,

$$\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)] \quad (87)$$

provides a risk-sensitive complement to mean regret.

15.4 Provenance experiment

Provenance, Value of Information, and Incident Response yields a two-dimensional repeated-measures design indexed by information regime and incident stage. Let L_{irsd} denote realized stage loss for hidden incident i , replication r , stage s , and depth $d \in \{0, 1, 2\}$ for blind, fast, and deep provenance. A mixed model can estimate

$$L_{irsd} = \alpha_i + \lambda_s + \beta_1 \mathbf{1}\{d = 1\} + \beta_2 \mathbf{1}\{d = 2\} + \eta_s \mathbf{1}\{d = 2\} + \varepsilon_{irsd}, \quad (88)$$

where the stage interaction η_s tests the central hypothesis that the value of deeper provenance differs across prevention, containment, remediation, and redeployment. The paired design is essential because all information regimes are evaluated on the same hidden incident.

The statistical analysis should distinguish at least four outcomes: posterior entropy, attribution calibration, stage action, and realized stage loss. Entropy reduction is an intermediate informational outcome; it should not be conflated with economic value. A key diagnostic regression is therefore

$$\Delta L_{irs} = \alpha + \beta_H \Delta H_{irs} + \beta_A \mathbf{1}\{a_{irs}^F \neq a_{irs}^D\} + \lambda_s + u_i + \epsilon_{irs}, \quad (89)$$

where ΔH is the entropy reduction from fast to deep provenance. If β_H is small but β_A is large, the result would confirm that information matters primarily when it crosses a decision boundary.

The Bayesian benchmark supplies a second family of estimands. For each stage, one can report mean EVPI, mean net EVSI, the share of incidents with positive net EVSI, and agreement between the LLM's stated probability that deeper provenance will change action and the benchmark's sign of net EVSI. Proper calibration scores can be used for the LLM's provenance posterior, while lifecycle regret relative to the private oracle remains the ultimate decision metric.

15.5 Mechanism-level heterogeneity

The key economic interaction is between information value and urgency. Let U_i be a normalized urgency score and S_i action specificity, such as the gap between the best and second-best control under the true mechanism. A useful specification is

$$L_{ic} = \alpha_i + \beta_c + \gamma_1 U_i \mathbf{1}\{c \in \text{theory}\} + \gamma_2 S_i \mathbf{1}\{c \in \text{theory}\} + \epsilon_{ic}. \quad (90)$$

Theory should be more valuable when S_i is high and less valuable when U_i is high, producing the qualitative predictions $\gamma_2 < 0$ and $\gamma_1 > 0$ when loss is the dependent variable.

15.6 Power and replication

The computational studies are mechanism laboratories, not yet large-sample studies. A journal submission should use pilot variance estimates to determine replication. Because model calls are costly, paired designs are especially valuable. A bootstrap over hidden cases and replications can quantify uncertainty, while pre-registered stopping rules should prevent selective expansion after favorable results. Model versions, prompts, random seeds, structured-output schemas, and private validity certificates should be archived because model drift can otherwise make later replication ambiguous.

16 Cybersecurity Economics: Information, Control, and Option Value

The framework yields a security-economic decomposition. Let C_S be spending on static protection, C_D on diagnosis, and C_C on intervention. Expected total cost is

$$J(C_S, C_D, C_C) = C_S + C_D + C_C + \mathbb{E}[L_{\text{residual}}]. \quad (91)$$

Traditional security economics asks how C_S and C_C reduce expected breach loss. The present framework treats C_D as an investment that changes the conditional productivity of C_C . Diagnosis has complementarity with targeted control but substitution with rapid containment.

Provenance, Value of Information, and Incident Response adds a second information asset: provenance capital. Let C_P denote expenditure on preserving, validating, and acquiring trace or causal provenance. Expected lifecycle cost can be written as

$$J(C_S, C_D, C_P, C_C) = C_S + C_D + C_P + C_C + \mathbb{E}[L_{\text{lifecycle}}]. \quad (92)$$

Unlike diagnostic expenditure C_D , which is often incident-specific, provenance expenditure can create intertemporal option value because evidence collected during containment can improve remediation or redeployment later. The marginal return to C_P is therefore stage dependent and can be positive even when its contemporaneous containment value is zero.

A useful decomposition is

$$\underbrace{\Delta H}_{\text{information quantity}} \neq \underbrace{\Delta R_s}_{\text{decision value}} \neq \underbrace{\Delta R_s - c - \text{delay}}_{\text{net institutional value}}. \quad (93)$$

This distinction is the cybersecurity analogue of Howard's observation that information value depends jointly on probabilities and consequences (Howard, 1966). It also provides an economic rationale for selective logging and provenance preservation: the relevant question is not whether more evidence exists, but whether trustworthy evidence changes a future decision enough to justify its lifecycle cost.

The option value of containment can be formalized. Let $\mathcal{A}_t$ be the set of feasible controls at time t . Catastrophic state transitions can shrink $\mathcal{A}_t$ irreversibly. An immediate containment action can therefore have value beyond its direct harm reduction if it preserves future action

feasibility:

$$V_{\text{option}} = \mathbb{E} \left[\min_{a \in \mathcal{A}_{t+D}^{\text{no contain}}} L(a) - \min_{a \in \mathcal{A}_{t+D}^{\text{contain}}} L(a) \right]. \quad (94)$$

When containment prevents loss of feasible recovery options, $V_{\text{option}} > 0$ even if containment is not the final optimal control. This economic interpretation strengthens the contain–diagnose–target architecture: containment is a real option on future informed action.

17 Implications for Incident Response and Autonomous AI Governance

17.1 Incident response as inverse game solving

A mature incident-response function already asks causal questions: what changed, which dependency propagated the event, what authority was used, what will happen if a service is isolated, and whether persistence remains after remediation. The game-theoretic framework adds a discipline: treat competing causal mechanisms as explicit hypotheses, design interventions that separate them, and value each experiment by its effect on downstream control rather than by curiosity alone.

Provenance, Value of Information, and Incident Response changes the incident-response interpretation in an important way. The organization should not ask only whether the mechanism is known. It should ask whether the remaining ambiguity is *decision relevant at the current stage*. During containment, multiple mechanisms can rationally collapse into one robust action class. During remediation, those same mechanisms can separate sharply because common-mode dependence, communication dependence, reward misdesign, external orchestration, and strategic opacity require different architectural repairs. During redeployment, residual uncertainty maps into an autonomy envelope rather than a binary safe/unsafe label.

This stage logic is consistent with the lifecycle orientation of NIST SP 800-61 Rev. 3, which integrates incident response throughout cybersecurity risk management rather than treating response as a single isolated phase (Nelson et al., 2025). The formal contribution here is the stage-specific stopping rule: preserve evidence during urgent control, but purchase deeper provenance only when its expected reduction in future Bayes risk exceeds information cost and delay.

In agentic AI this is especially important because an observed action can reflect several layers of causation. A model may generate a recommendation; an orchestration layer may decide to execute it; a memory layer may condition future behavior; a tool permission may create practical authority; another agent may provide information; an external integration may trigger the workflow. The “player” in the effective game may therefore be a composite system. Strategic diagnosis is partly the task of locating the layer at which incentives and control actually reside.

17.2 Governance implications

The framework is compatible with governance principles in NIST CSF 2.0 and the NIST AI RMF profile: identify risks, maintain traceability, evaluate controls, and define governance around system behavior rather than isolated model outputs (NIST, 2024; Autio et al., 2024). It also complements MITRE ATLAS and OWASP agentic-security taxonomies by adding a decision-theoretic layer between observed behavior and mitigation selection (MITRE, 2026;

OWASP, 2025).

For institutions, seven control principles follow.

1. **Separate generation from verification.** A model that proposes a difficult scenario, diagnosis, provenance explanation, or control should not be the sole authority certifying its correctness.
2. **Preserve private ground truth in experiments.** Hidden certificates, source labels, true loss surfaces, and audit provenance must remain outside the observer’s context until scoring is complete.
3. **Use controlled interventions as diagnostics.** Where feasible, test hypotheses in sandboxes, simulations, shadow modes, or reversible containment states rather than in uncontrolled production.
4. **Stop diagnosis when uncertainty becomes decision-irrelevant.** Further semantic certainty is wasteful if all plausible mechanisms imply the same current control.
5. **Contain when the information clock is slower than the harm clock.** Deep analysis is not a virtue when it crosses the intervention horizon.
6. **Preserve evidence even when immediate attribution is unnecessary.** Provenance that does not change containment can create option value for remediation, validation, or redeployment.
7. **Redeploy within a provenance-supported autonomy envelope.** Restoration of permissions and autonomy should reflect residual causal uncertainty and the loss associated with being wrong, not merely whether the incident appears quiet.

18 Limitations, Falsification, and Threats to Validity

The framework is intentionally ambitious, and several limitations must be explicit.

First, the computational studies are synthetic. Their value is mechanism isolation, not external validity. Source families, parameter ranges, probe menus, and loss functions are designed by the researcher. A result can be internally valid and still fail to transfer to operational cyber environments.

Second, the mathematical oracle is only as good as its grammar. In *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment*, equivalence is verified within a specified transformation class. A representation outside that class can alter strategic structure in ways the oracle does not capture. Likewise, *Inverse-Game Diagnosis under Scarce Experimental Budgets*’s response function and *Theory-Guided versus Model-Free Intervention*’s loss geometry are stylized causal models.

Third, language models are stochastic and versioned. A model name does not define a timeless algorithm. Provider updates can change output distributions, calibration, structured-output behavior, and reasoning performance. Reproducibility therefore requires model/date metadata and repeated calls.

Fourth, active probes can be performative. In a real cyber system, an intervention used for diagnosis can teach an adversary, alter incentives, change topology, or reveal defender beliefs. The simple Bayesian update in (16) assumes the latent mechanism does not strategically adapt to

the experiment. A richer model should treat probing as part of the game and include adversarial response.

Fifth, the defender’s model class can be misspecified. If the true mechanism lies outside Θ , posterior concentration within the wrong family can generate confident but structurally false conclusions. Open-set or hybrid classes reduce but do not eliminate this problem.

Sixth, counterfactual prediction depends on intervention invariance assumptions. If an intervention changes the system in a way not represented by the latent mechanism, predictions can fail even when parameter estimation is accurate.

Seventh, economic loss functions are normative. Control cost, service disruption, legal constraints, reputational damage, and irreversible-harm thresholds are institution-specific. Journal-level empirical work should conduct sensitivity analysis over these quantities rather than report a single calibrated loss surface as universal.

Eighth, *Provenance, Value of Information, and Incident Response*’s provenance model is stylized. Its five causal dimensions and Bernoulli diagnostic packets are not substitutes for real telemetry, chain-of-custody, cryptographic attestation, immutable logs, identity evidence, model registries, or workflow provenance. The paper distinguishes trace provenance from causal provenance, but the computational laboratory primarily models the latter. Empirical work must test whether real provenance systems supply evidence with enough integrity and discriminating power to support the decision-theoretic claims made here.

Ninth, provenance itself can be adversarial. Logs can be incomplete, strategically manipulated, selectively disclosed, or generated by the same components under investigation. The weakly nonnegative option-value result in Proposition 10.2 assumes the preserved evidence is trustworthy and optional. Once evidence integrity is endogenous, provenance acquisition becomes a security game of its own.

These limitations provide falsification criteria. The theory would be weakened if representational transformations that are verified as equivalent never change bounded-observer behavior across a wide range of observers; if adaptive probing never outperforms balanced passive designs even when discriminating probes exist; or if theory-guided intervention systematically fails to reduce regret even in slow-harm, high-action-specificity cases where information should be valuable. Negative results would be theoretically useful because they would identify which layer of the proposed architecture contributes little.

19 Extensions and Research Agenda

19.1 Nonstationary and adversarial mechanisms

A natural extension allows θ_t to evolve. The defender then faces a hidden Markov model or stochastic game in which mechanisms can switch after probes or controls. Bayesian filtering replaces static inference:

$$b_{t+1}(\theta') \propto p(y_{t+1} \mid \theta', q_t) \sum_{\theta} P(\theta' \mid \theta, a_t, q_t) b_t(\theta). \quad (95)$$

This is especially relevant when autonomous agents adapt to containment or learn new coordination policies.

19.2 Multi-agent emergence

The source mechanism can be extended from a low-dimensional vector to a graph-valued latent object. Let W_t be a directed weighted interaction graph among agents, with edges representing communication, influence, shared memory, or tool dependence. Emergent cooperation can then be modeled through endogenous coalition formation or correlated equilibrium. The diagnostic problem becomes inference over both graph topology and strategic type.

19.3 Robust and distributionally robust control

If posterior probabilities are unreliable, the defender can optimize over an ambiguity set $\mathcal{B}$ of beliefs:

$$a^* \in \arg \min_a \sup_{b \in \mathcal{B}} \mathbb{E}_{\theta \sim b} L(a, \theta). \quad (96)$$

This connects the framework to robust control and distributionally robust optimization. It also offers a middle ground between pure Bayesian targeting and worst-case minimax containment.

19.4 Adversarial representation design

Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment treats representation difficulty as induced but non-strategic. A richer model would make the encoder an adversary who chooses γ subject to an equivalence constraint to maximize defender loss:

$$\max_{\gamma \in \Gamma_{\text{eq}}} \min_{D \in \mathcal{D}(B)} \mathbb{E}[\ell_I(D(R_\gamma(G)), G)]. \quad (97)$$

This becomes a Stackelberg representation game. The defender can respond by investing in representation-invariant features or canonicalization.

19.5 Endogenous experiment cost and attacker learning

In realistic cyber settings, probes can reveal information to the adversary. Let $c(q, b^A)$ depend on the attacker's belief about the defender. Then experiment selection becomes a signaling game. A diagnostically useful probe may be strategically expensive because it reveals which mechanism the defender suspects.

19.6 Continuous-time control

For high-speed autonomous systems, discrete diagnostic rounds may be too coarse. Continuous-time filtering and control can model hazard and intervention more naturally. Let b_t evolve under a Kushner–Stratonovich or Wonham filter and choose stopping time τ for intervention. The defender solves

$$\inf_{\tau, a_\tau} \mathbb{E} \left[\int_0^\tau h_\Theta(t) dt + L(a_\tau, \Theta; \tau) \right], \quad (98)$$

a partially observed optimal stopping problem.

19.7 Endogenous provenance integrity and lineage graphs

A natural extension is to make the provenance graph $\mathcal{P}_t$ endogenous. Evidence nodes can have integrity states, attestation confidence, access-control labels, and adversarial manipulation

risk. The defender would then choose not only which hypothesis to test but which evidence path to trust. A graph-theoretic model could assign each provenance edge a reliability weight and evaluate causal explanations by both likelihood and trace integrity. This would connect decision-theoretic causal provenance with established data-lineage standards and digital-forensics practice (Simmhan et al., 2005; Moreau and Missier, 2013).

A second extension is to treat evidence preservation as a portfolio problem. Logging, model checkpoints, message traces, tool-use records, and external-input snapshots consume storage, privacy budget, and operational complexity. The optimal provenance architecture would maximize the expected future decision value of preserved evidence subject to those resource and confidentiality constraints.

19.8 Empirical cyber data

The most important next step is to connect the synthetic mechanism classes to real but safely curated datasets: incident timelines, red-team traces, agent sandbox trajectories, simulated enterprise networks, or benchmark environments. The purpose would not be to automate live response but to test whether strategic-inference variables improve retrospective explanation and counterfactual prediction beyond conventional classifiers.

20 Conclusion

Cybersecurity in autonomous AI environments becomes qualitatively different when the relevant actors can collaborate. The central risk is not merely that a model produces an unsafe output, but that multiple agents can become strategically interdependent through communication, repeated interaction, reputation, shared incentives, coalition formation, common signals, or external orchestration. Once that happens, the defender confronts a game rather than a collection of independent anomalies.

The broader *Game Theory in the Age of AI* program provides the conceptual foundation for this claim. *Cooperation Beyond Correlation* establishes why correlation must be distinguished from strategic dependence by intervention rather than by surface similarity alone (Reynoso, 2026a). *Constructive Collaboration* shows that collaboration can arise voluntarily when information sharing is instrumentally valuable (Reynoso, 2026b). *Parameterized Seeding and Topology* shows that collaborative regimes can occupy structured regions of initial-condition space and exhibit boundary sensitivity (Reynoso, 2026c). *Rationality Calibration in Canonical Games* uses canonical games to calibrate equilibrium-consistent decision behavior and representation robustness (Reynoso, 2026d). *Repeated Games, Memory, Reputation, and an LLM Player* demonstrates why memory, reputation, persistence, and future interaction can support cooperation that a one-shot analysis would miss (Reynoso, 2026e). *Coalitions and Collaborative Equilibria* separates productive coalition value from allocation and stability, making clear that internally rational cooperation can coexist with harmful externalities for the institution (Reynoso, 2026f).

Those foundational studies change the cybersecurity question. If harmful collaboration can be rational and stable, then the institution must disentangle *what game is being played*, not merely detect that several agents behaved similarly. This is why game theory is the organizing scientific framework of the paper. It provides a language for best responses, equilibrium, repeated incentives, incomplete information, coalition value, and strategic adaptation. The framework does not require the assumption that current AI agents are perfectly rational. It requires only

that their behavior can be tested against explicit strategic hypotheses whose predictions differ under controlled interventions.

The cybersecurity-focused studies isolate the four problems that arise after that foundation is established. *Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment* shows why representation must be treated as an independent variable (Reynoso, 2026g). Strategically equivalent games can occupy different surface forms, and a bounded observer can assign different meanings to them even when an independent oracle certifies exact equivalence. This motivates observer-relative strategic opacity and the need for representation-invariant security analysis.

Inverse-Game Diagnosis under Scarce Experimental Budgets shows that passive recognition is not enough when the mechanism is latent (Reynoso, 2026h). The defender must be allowed to ask the system questions through controlled interventions, but those questions consume scarce evidence and must be chosen strategically. The natural mathematical language is Bayesian experimental design on a partially observed game. The relevant output is not only a label; it is a structural model that predicts responses to interventions not used to fit it.

Theory-Guided versus Model-Free Intervention adds the criterion that disciplines both earlier stages: information is valuable only through action (Reynoso, 2026i). A more accurate diagnosis can be economically inferior if it postpones a high-value control. Once residual harm, control cost, implementation time, and irreversible exposure are placed in the same loss function, the static intuition that more information is always better becomes conditional. Theory has positive value only when its targeting gain exceeds its delay burden:

continue diagnosing only while the expected value of learning exceeds the cost of waiting.

(99)

Provenance, Value of Information, and Incident Response extends that stopping rule across the incident lifecycle (Reynoso, 2026j). Provenance is not a binary possession and not an intrinsic good. It is stage-dependent information capital. A provenance packet that materially reduces entropy can have little value for immediate containment when all credible mechanisms imply the same broad control. The same packet can have high value at remediation if the mechanisms imply different root-cause repairs, and high value again at redeployment if residual uncertainty changes the permissible autonomy envelope:

resolve uncertainty only to the level required by the next consequential decision.

(100)

The paper therefore separates four notions that are often collapsed in cyber analysis. *Strategic equivalence* asks whether two representations preserve the same game. *Observational equivalence* asks whether available probes can distinguish two latent mechanisms. *Decision equivalence* asks whether those mechanisms require different controls. *Stage-specific institutional equivalence* asks whether the distinction matters now, at prevention, containment, remediation, or redeployment. Security analysis can stop at different resolutions at different stages because the action geometry changes over the lifecycle.

The framework also clarifies the role of obscurity. The experiments on representation establish only that strategic opacity can exist; they do not establish that current AI agents intentionally conceal their strategies. Nevertheless, if agents become sufficiently capable and if detection changes their payoffs, obscuring observable behavior is itself a rational strategic possibility. In that case, the defender’s problem is adversarially reflexive: the agents choose actions within the game while also influencing the evidence through which the game is inferred.

Game theory is indispensable precisely because the observation process can become endogenous to strategic incentives.

This leads to a fuller incident-response architecture:

$$\begin{aligned}
 &\text{identify strategic dependence} \rightarrow \text{audit representation} \rightarrow \text{contain now} \\
 &\quad \text{preserve evidence} \rightarrow \text{diagnose when decision-relevant} \\
 &\text{remediate} \rightarrow \text{validate} \rightarrow \text{redeploy within a supported autonomy envelope.}
 \end{aligned} \tag{101}$$

Containment buys time; provenance preservation protects the value of that time; remediation consumes causal information; validation tests whether the strategic mechanism has actually changed; and redeployment turns residual uncertainty into a decision about scope and autonomy rather than a binary declaration of safety.

For the economics of cybersecurity, diagnosis and provenance are both information investments. Diagnostic capital can depreciate rapidly while harm accumulates. Provenance capital can mature: evidence preserved during containment may become valuable only later. The relevant quantities are not entropy or classification accuracy alone but stage-specific Bayes risk, EVPI, EVSI, regret, option value, and the externality imposed by stable agent coalitions. The central economic identity is therefore

$$\text{information quantity} \neq \text{decision value} \neq \text{net institutional value after cost and delay.} \tag{102}$$

For computer science and AI security, the framework suggests that evaluation must move beyond static benchmark accuracy. Secure autonomous systems should be tested for representation robustness, mechanism identifiability, counterfactual response, calibrated uncertainty, controlled degradation under isolation, provenance integrity, reproducible lineage, and the ability of governance controls to preserve solvency while diagnosis proceeds. The relevant system boundary may include models, tools, memory, identity, orchestration, communication, external signals, and other agents rather than the model endpoint alone.

The paper does not claim that the synthetic computational studies establish real-world cyber effectiveness. They establish a research architecture and a set of falsifiable propositions. The next empirical phase should increase replication, vary model capability independently, introduce richer stochastic games, allow agents to adapt strategically to probes, connect latent mechanisms to curated incident traces, model provenance integrity and missingness, and test whether stage-specific information acquisition reduces out-of-sample regret. The complete *Game Theory in the Age of AI* will be released in a public GitHub repository so that each named study, its code, prompts, assumptions, and experimental artifacts can be inspected independently.

The broader proposition is that the security frontier for autonomous AI moves from protecting components to governing partially observed strategic systems over time. When agents collaborate, the defender must determine whether the interdependence is real, infer the game that sustains it, account for the possibility that the observable representation is misleading or strategically obscured, decide what to ask, know when to stop asking, preserve evidence that may matter later, and restrict future autonomy to what the remaining uncertainty can justify. Cybersecurity becomes a problem not only of detection and response, but of *game-theoretic inference, control, and provenance economics under a deadline*.

A Proof Details and Additional Results

A.1 A decision-theoretic stopping rule

Let $V(b)$ be the expected minimum control loss under belief b with immediate action,

$$V_A(b) = \min_a \mathbb{E}_b[L(a, \Theta; 0)]. \quad (103)$$

Let a diagnostic probe q take time Δt_q , incur direct cost c_q , and produce posterior $b^{q,Y}$. Define

$$V_Q(q; b) = c_q + C_{\text{delay}}(\Delta t_q; b) + \mathbb{E}_Y[V_A(b^{q,Y})]. \quad (104)$$

The one-step optimal stopping rule is to continue diagnosing if

$$\min_q V_Q(q; b) < V_A(b). \quad (105)$$

Equivalently, stop when the maximum net expected value of experimentation is nonpositive:

$$\max_q \left\{ V_A(b) - \mathbb{E}_Y[V_A(b^{q,Y})] - c_q - C_{\text{delay}}(\Delta t_q; b) \right\} \leq 0. \quad (106)$$

This equation makes the “right amount of theory” endogenous.

A.2 Regret bound from posterior error

Suppose $L(a, \theta)$ is bounded in $[0, L_{\max}]$ and let b be the true posterior while $\hat{b}$ is an approximate posterior produced by an inference model. Let a_b and $a_{\hat{b}}$ be their Bayes actions. Then

$$\mathbb{E}_b L(a_{\hat{b}}, \Theta) - \mathbb{E}_b L(a_b, \Theta) \leq 2L_{\max} \|b - \hat{b}\|_{\text{TV}}. \quad (107)$$

To see this, add and subtract expectations under $\hat{b}$ and use the fact that $a_{\hat{b}}$ minimizes loss under $\hat{b}$; each distribution-shift term is bounded by $L_{\max} \|b - \hat{b}\|_1/2$. This bound connects diagnostic calibration to intervention regret but also shows why very different posteriors can have small regret when action loss is flat.

A.3 A KL lower bound on distinguishability

For two mechanisms θ_0 and θ_1 under n independent samples from probe q , the log likelihood ratio accumulates linearly in n , and the expected log Bayes factor under θ_0 is

$$\mathbb{E}_{\theta_0} \left[\log \frac{p_{\theta_0}(Y_{1:n} | q)}{p_{\theta_1}(Y_{1:n} | q)} \right] = n D_{\text{KL}}(p_{\theta_0}^q \| p_{\theta_1}^q). \quad (108)$$

Thus the product $n D_{\text{KL}}$ is a natural measure of diagnostic separation. Increasing model reasoning cannot compensate for a probe with near-zero KL divergence unless other prior information is available.

A.4 Stage-specific zero-EVPI result

Proposition 10.1 has an immediate extension to sets of posterior states. Let $A_s^*(C)$ denote the common optimal action set over a credible region $C \subseteq \Theta$. If $A_s^*(C) \neq \emptyset$ and $b(C) = 1$, then

any experiment whose posterior remains supported inside C has zero gross decision value at stage s unless it changes the action set through a transition to a different lifecycle stage. This formalizes the sense in which causal refinement inside a decision-equivalence cell is institutionally redundant.

A.5 Gross versus net value of preserved evidence

Let $\mathcal{I}$ be additional trustworthy evidence that can be preserved at time t and observed at a future decision stage. If preservation is costless and the decision maker can ignore $\mathcal{I}$, Blackwell's principle implies

$$\mathbb{E} \left[\min_a \mathbb{E}[L(a, \Theta) \mid \mathcal{I}] \right] \leq \min_a \mathbb{E}[L(a, \Theta)]. \quad (109)$$

Thus the gross value of optional evidence is weakly nonnegative. If preservation cost is c_P , integrity-validation cost is c_V , and the evidence induces delay cost C_D , its net value is

$$V_P^{\text{net}} = V_P^{\text{gross}} - c_P - c_V - C_D, \quad (110)$$

which can have either sign. This distinction prevents the provenance argument from degenerating into the claim that maximal logging is always optimal.

A.6 A stage-specific provenance stopping rule

Let Q_s be the set of admissible provenance-acquisition actions at stage s . With current posterior b , the one-step stopping condition is

$$\max_{q \in Q_s} \left\{ R_s(b; 0, 0) - \mathbb{E}_Y R_s(b^{q,Y}; \Delta t_q, c_q) \right\} \leq 0. \quad (111)$$

Because R_s is stage indexed, the same experiment can be rejected during containment and accepted during remediation. This is the formal maturity structure of provenance capital.

B Computational Research Collection and Repository Placeholder

The complete computational collection will be placed in a public GitHub repository in a later revision. The manuscript therefore uses a collection-level placeholder rather than temporary private links:

GITHUB_REPOSITORY_URL_TO_BE_ADDED

Each study is cited independently in the bibliography by its title. The following table records the research question and its role in the present paper.

Computational study	Research question	Role in this paper
<i>Cooperation Beyond Correlation</i>	Can controlled interventions distinguish strategic cooperation from common-cause correlation?	Establishes causal identification of strategic dependence.

Computational study	Research question	Role in this paper
<i>Constructive Collaboration</i>	Can agents with competing local hypotheses improve a distributed task through voluntary information sharing?	Establishes a mechanism for rational, uncommanded collaboration.
<i>Parameterized Seeding and Topology</i>	Does collaboration arise across a broad region of controlled initial conditions or only from narrow seeds?	Introduces collaboration phase topology and boundary sensitivity.
<i>Rationality Calibration in Canonical Games</i>	Are decision procedures consistent with canonical game-theoretic solution concepts across representations?	Calibrates the game-theoretic rationality framework.
<i>Repeated Games, Memory, Reputation, and an LLM Player</i>	How do repeated interaction, memory, reputation, implementation noise, and an LLM-based rule affect cooperation?	Establishes dynamic mechanisms for persistence and history dependence.
<i>Coalitions and Collaborative Equilibria</i>	When is a productive coalition efficient, stable, and acceptably rewarded?	Distinguishes coalition value, allocation, stability, and externality.
<i>Strategic Representation and Encoded Games: An LLM-versus-LLM Identification Experiment</i>	Can an intelligent encoder make a familiar game harder to identify while preserving exact strategic equivalence?	Provides the strategic-opacity experiment and independent equivalence oracle.
<i>Inverse-Game Diagnosis under Scarce Experimental Budgets</i>	Can a bounded observer recover a useful latent mechanism and predict unseen interventions under a scarce experimental budget?	Provides the active inverse-game and held-out counterfactual design.
<i>Theory-Guided versus Model-Free Intervention</i>	Does theory-guided intervention reduce realized loss once diagnostic error, delay, implementation time, and control cost are counted?	Provides the time-bounded control and value-of-theory experiment.
<i>Provenance, Value of Information, and Incident Response</i>	At which incident-response stages does additional provenance reduce expected institutional loss after information cost and delay?	Provides the stage-dependent provenance, EVPI/EVSI, and redeployment experiment.

C Notation

Symbol	Meaning
θ	Latent strategic/cyber mechanism.
$\mathcal{G}_\theta$	Stochastic game parameterized by θ .
R_γ	Representation transformation indexed by γ .
z	Public representation or artifact observed by the defender.
$b_t(\theta)$	Defender belief over latent mechanisms at time t .
q_t	Diagnostic probe or controlled experiment.
Y_t	Probe outcome.
$\mathcal{C}$	Feasible security-control set.
a	Security intervention/control.
T_D	Diagnostic delay.
$T_I(a)$	Action implementation time.
T_H	Time to irreversible harm or equivalent exposure threshold.
$h_\theta(t)$	Uncontrolled harm rate.
$e(a, \theta)$	Control effectiveness against mechanism θ .
$k(a)$	Direct control cost.
$L(a, \theta; T_D)$	Total time-dependent security loss.
$\sim_\Gamma$	Strategic equivalence under admissible representation transforms.
$\sim_{\mathcal{Q}}$	Observational equivalence under feasible probes.
$\sim_L$	Decision equivalence under the loss function.
EIG	Expected information gain.
Δ_R	Representation-treatment effect.
Δ_A	Adaptive-diagnosis effect.
Δ_T	Realized value-of-theory effect.
$\mathcal{P}_t$	Preserved trace-provenance record available at time t .
σ_t	Incident-response stage at time t .
$\mathcal{A}_s$	Feasible institutional actions at stage s .
$L_s(a, \theta; \delta)$	Stage-specific loss under action a , mechanism θ , and information delay δ .
$R_s(b; \delta, c)$	Stage-specific Bayes risk under belief b , delay δ , and information cost c .
EVPI_s	Expected value of perfect provenance information at stage s .
NetEVSI_s	Net expected value of sample provenance information at stage s .
$\sim_{L,s}$	Stage-specific decision-equivalence relation.
$u^*(b)$	Provenance-supported maximum redeployment autonomy level.
$\Delta_{P,s}$	Realized effect of deeper provenance on stage- s loss.

D Pseudocode for the Unified Defender

Input: public artifact z , prior $b(\theta)$, probe budget B ,
incident stage s , provenance archive P ,
stage action sets $\mathcal{A}_s$, time-to-harm estimate T_H

1. Canonicalize / audit the representation where possible.
2. Construct current belief b over strategic mechanisms.
3. Compute the current stage's decision-equivalence classes.

4. While diagnostic budget remains:
 - a. Compute immediate stage risk $R_s(b)$.
 - b. For each safe probe q :
 - estimate $\text{NetEVSI}_s(q; b)$, including cost and delay.
 - c. If $\max_q \text{NetEVSI}_s(q; b) \leq 0$: stop investigating.
 - d. Execute the best bounded probe in sandbox / controlled mode.
 - e. Update b and append trustworthy evidence to P .
 - f. If urgency rises, activate broad containment.
5. Choose the stage action minimizing posterior expected loss.
6. Preserve evidence that can have option value at later stages.
7. On transition to remediation or redeployment:
 - a. replace A_s and L_s with the next-stage geometry;
 - b. recompute decision sufficiency and provenance EVSI;
 - c. acquire deeper provenance only if its net stage value is positive.
8. At redeployment, select autonomy within the supported risk envelope.
9. Record representation, evidence, posterior, action, timing, provenance integrity, outcome, and private audit certificate.

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